Resource scaling of the quantum-linear-systems pathway for the viscous Burgers equation

An error budget and complexity analysis of the block-encoding + QSVT method, with comparison to classical solvers and to Koopman–von Neumann linearisation

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Abstract

Setty [*J. Phys. A: Math. Theor.* **59**, 185303 (2026)] proposes a pathway that solves differential equations on a quantum computer by discretising them into a linear system $\mathsf{L}\boldsymbol{y} = \boldsymbol{b}$, block-encoding L, and inverting it with the quantum singular value transformation (QSVT); nonlinear equations are brought into this form by Carleman linearisation. That work demonstrates the pathway at circuit level on a 30×30 instance of the viscous Burgers equation and reports two-qubit gate depths, but it does not derive how the cost scales. This report supplies that analysis. We enumerate every error source in the pipeline, show which of them are amplified by the condition number, and derive the qubit count, the block-encoding subnormalisation, the polynomial degree, the circuit depth, the post-selection success probability and the sample complexity as explicit functions of the target error ϵ, the Reynolds number Re, the grid size and the Carleman order. Every formula is validated numerically against the published instance; our success-probability model reproduces the reported value to 5% and our subnormalisation formula is exact. Four findings stand out. (i) The quantity that sets the QSVT polynomial degree is $\alpha/\sigma_{\min}$, not $\sigma_{\max}/\sigma_{\min}$; the two differ by a factor of 3.5 in the published instance. (ii) With the coherent-permutation block encoding used in the paper, the subnormalisation grows as n_x^{N-1} with grid size n_x and Carleman order N, which erases the nominal exponential compression; replacing it with an arithmetic sparse-access oracle restores an $\mathcal{O}(N^2)$ scaling. (iii) Solving one time step at a time, as the paper does, makes the success probability decay as p_s^m over m steps and is not viable; a dilated "history-state" system is required, and we show that the product of subnormalisation and step count entering its condition number is independent of the time step. (iv) The Carleman convergence criterion evaluates to $\mathrm{Re} < \pi/2$ for this discretisation, so the published instance ($\mathrm{Re} = 100$) lies far outside the regime in which the linearisation is provably convergent. Assembling everything, the sequential gate depth scales as $\widetilde{\mathcal{O}}\big(\kappa_V\, T\, \mathrm{Re}^{-1}\epsilon^{-1} N^5\big)$ against a classical pseudo-spectral cost of $\widetilde{\mathcal{O}}(T\epsilon^{-1})$ arithmetic operations: in one space dimension there is no speedup of any kind. We close by explaining what changes for Koopman–von Neumann linearisation — the derivation transfers almost verbatim, but the state space grows as M^{n_x} rather than n_x^N — and by listing the specific changes that would be needed for the pathway to become competitive.

Contents

1 Introduction and scope

1.1 What problem this report addresses

Quantum computers evolve states linearly. A nonlinear partial differential equation (PDE) therefore cannot be simulated directly; it must first be *embedded* in a linear problem, and only then handed to a quantum subroutine. The pathway analysed here, due to Setty [1], chains together four such steps for the one-dimensional viscous Burgers equation:

$$\text{PDE} \xrightarrow{\text{finite differences}} \text{quadratic ODE system} \xrightarrow{\text{Carleman}} \text{linear ODE system}$$
$$\xrightarrow{\text{implicit time stepping}} \text{linear algebraic system} \xrightarrow{\text{block encoding + QSVT}} \text{quantum state}$$

Reference [1] establishes that this chain *works*: it constructs the explicit gate-level circuit for a 30×30 Burgers instance, runs it in simulation, and reports the resulting two-qubit gate depths on two IBM processor topologies. What it does not do — and what the present report supplies — is derive how each resource behaves as the instance is made larger and more accurate. Without that, one cannot say whether the pathway is a promising route to quantum advantage or a demonstration that happens to fit on eleven qubits.

The analysis is written at the level of a master's student in physics. Every symbol is defined before it is used, and every piece of background that goes beyond a first course in numerical methods and quantum computing is developed in an appendix rather than cited. Section 2.3 collects all notation in one table for reference.

1.2 What is assumed known, and what is developed here

We assume the reader is comfortable with linear algebra (eigenvalues, singular values, matrix norms), with the idea of a finite-difference approximation to a derivative, and with the basic formalism of quantum computing (qubits, unitary gates, measurement, the Dirac notation $|\psi\rangle$). Everything else is built up:

- Appendix A — finite-difference stencils and their truncation errors, including the Gronwall argument that converts a local error into a global one.
- Appendix B — the Kronecker product algebra needed to manipulate the Carleman blocks.
- Appendix C — exact spectra of the discrete diffusion and Carleman operators.
- Appendix D — quantum signal processing, QSVT, and the polynomial approximation of $1/x$.
- Appendix E — what a block encoding is, and the two concrete constructions we compare.
- Appendix F — amplitude amplification and amplitude estimation, and why they enter the sample complexity.
- Appendix G — the perturbation bound that explains why some errors get multiplied by the condition number and others do not.
- Appendix H — the Cole–Hopf transformation, which is the reason Burgers is an unrepresentative test case.
- Appendix I — the numerical scripts used to validate every formula in the report.

1.3 Summary of findings

For the reader who wants the conclusions before the derivations:

1. **The relevant condition number is $\alpha/\sigma_{\min}$.** The QSVT polynomial degree is set not by the classical condition number $\sigma_{\max}/\sigma_{\min}$ of the linear system, but by the ratio of the block-encoding subnormalisation α to the smallest singular value. For the published Burgers instance these are 5.46 and 1.57 respectively — a factor 3.5 apart. The former matches the design value $\kappa = 8$ used in [1]; the latter does not (Section 5.2).

2. **The block encoding, not the matrix, is the bottleneck.** For the coherent-permutation block encoding of [1, 2] we derive and numerically confirm $\alpha \simeq n_x^{N-1}\lambda_{\text{cfl}}$, where n_x is the number of interior grid points, N the Carleman order and $\lambda_{\text{cfl}} = \Delta t/\Delta x$ the Courant number. Because α enters the cost linearly, the algorithm is *polynomial in the grid size and exponential in the Carleman order* — exactly the scaling the Carleman embedding was supposed to avoid. An arithmetic sparse-access oracle instead gives $\alpha = \mathcal{O}(N^2\lambda_{\text{diff}})$, independent of n_x except through the diffusion number (Section 5.3).

3. **Sequential time stepping does not scale.** Reference [1] solves one implicit step at a time. Because each solve is heralded with probability $p_s < 1$ and the output of one step is the input of the next, chaining m steps costs $\mathcal{O}(p_s^{-m/2})$ even with optimal amplitude amplification. The dilated "history-state" formulation removes this, and we show that the product αm entering its condition number is *independent of* Δt, which is what makes the whole pathway tractable at all (Section 5.7).

4. **Carleman convergence requires** $\mathrm{Re} < \pi/2$**.** Evaluating the standard convergence ratio for this specific discretisation gives $R_\star = 2\mathrm{Re}/\pi$, so the Carleman series has a provable convergence regime only for $\mathrm{Re} \lesssim 1.6$. The instance in [1] has $\mathrm{Re} = 100$, giving $R_\star \approx 64$; we confirm numerically that the ratio computed directly from the matrices is 76. The published example is a circuit demonstration, not a converged simulation, and should not be read otherwise (Section 4.3).

5. **No speedup in one dimension.** Assembling all of the above, the quantum sequential depth is $\widetilde{\mathcal{O}}(\kappa_V T \mathrm{Re}^{-1}\epsilon^{-1}N^5)$ logical two-qubit gates, against $\widetilde{\mathcal{O}}(T\epsilon^{-1})$ floating-point operations for a classical pseudo-spectral solver. The ϵ scaling is identical, the quantum route carries extra factors of N^5 and of the eigenvector condition number κ_V, and it returns a quantum state rather than the field. A crossover can only be sought in $d \geq 3$ spatial dimensions with observable-only readout (Section 6).

Remark 1.1 (On tone). None of this is a criticism of [1], whose stated purpose is to consolidate the pathway and provide gate-level resource estimates for small instances — a genuinely useful contribution, and one whose published numbers we rely on throughout as calibration data. The scaling questions were simply left open, and answering them honestly is the point of this report.

2 Problem statement and notation

2.1 The equation and the deliverable

We study the one-dimensional viscous Burgers equation

$$\frac{\partial u}{\partial t} + u\,\frac{\partial u}{\partial x} = \nu\,\frac{\partial^2 u}{\partial x^2}, \qquad x \in [0, L], \quad t \in [0, T], \tag{1}$$

where
- $u = u(t, x) \in \mathbb{R}$ is the unknown scalar field (a velocity);
- $\nu > 0$ is the kinematic viscosity (units: length^2/time);
- $L > 0$ is the length of the spatial domain;
- $T > 0$ is the final time at which we want the solution.

The initial condition is $u(0, x) = u_0(x)$, taken smooth, with amplitude

$$U := \|u_0\|_\infty = \max_{x\in[0,L]} |u_0(x)|. \tag{2}$$

Reference [1] imposes homogeneous Dirichlet conditions $u(t, 0) = u(t, L) = 0$; the challenge specification uses periodic conditions $u(t, 0) = u(t, L)$. The two differ only in the boundary rows of the matrices below and change none of the scaling results, so we carry the Dirichlet case (to match the paper we are analysing) and flag the two places where periodicity would alter a constant.

The nonlinear term $u\,\partial_x u$ is what makes this hard: it is quadratic in u, and a quantum computer applies linear maps. The diffusive term $\nu\,\partial_{xx} u$ is linear and, crucially, *dissipative* — it damps the solution — which is what makes the Carleman embedding of Section 3.2 converge at all.

Deliverable. Following the challenge specification, we are given an oracle that prepares the amplitude encoding of the initial data,

$$\mathcal{U}_0 : \ |0\rangle \ \longmapsto \ \frac{1}{\|\boldsymbol{u}_0\|_2} \sum_{j=0}^{n_x-1} u_0(x_j)\,|j\rangle, \tag{3}$$

where x_j are the grid points defined in Section 3.1 and $\boldsymbol{u}_0 \in \mathbb{R}^{n_x}$ is the vector of initial values, and we must produce the amplitude encoding of a numerical approximation to $u(T, \cdot)$,

$$|\hat{u}(T)\rangle \ = \ \frac{1}{\|\hat{\boldsymbol{u}}(T)\|_2} \sum_{j=0}^{n_x-1} \hat{u}_j(T)\,|j\rangle, \tag{4}$$

together with honest estimates of the resources required. We write $\hat{\cdot}$ throughout for a numerically computed approximation, reserving unhatted symbols for exact quantities.

Accuracy target. We measure error in the relative discrete ℓ^2 norm,

$$\epsilon \ := \ \frac{\|\hat{\boldsymbol{u}}(T) - \boldsymbol{u}^{\text{exact}}(T)\|_2}{\|\boldsymbol{u}^{\text{exact}}(T)\|_2}, \qquad \|\boldsymbol{v}\|_2 := \Big(\textstyle\sum_j |v_j|^2\Big)^{1/2}, \tag{5}$$

where $\boldsymbol{u}^{\text{exact}}(T)$ denotes the exact PDE solution sampled on the grid. This is the natural norm here because it is the one the amplitude encoding (4) actually represents.

2.2 Non-dimensionalisation

It is much easier to read the scaling results if the problem carries only one physical parameter. Introduce dimensionless variables

$$\bar{x} = \frac{x}{L}, \qquad \bar{t} = \frac{tU}{L}, \qquad \bar{u} = \frac{u}{U}, \tag{6}$$

so that $\bar{x} \in [0, 1]$. Substituting into (1) and using the chain rule ($\partial_t = (U/L)\partial_{\bar{t}}$, $\partial_x = (1/L)\partial_{\bar{x}}$) gives

$$\frac{U^2}{L}\frac{\partial \bar{u}}{\partial \bar{t}} + \frac{U^2}{L}\bar{u}\frac{\partial \bar{u}}{\partial \bar{x}} = \frac{\nu U}{L^2}\frac{\partial^2 \bar{u}}{\partial \bar{x}^2} \quad \Longleftrightarrow \quad \frac{\partial \bar{u}}{\partial \bar{t}} + \bar{u}\frac{\partial \bar{u}}{\partial \bar{x}} = \frac{1}{\mathrm{Re}}\frac{\partial^2 \bar{u}}{\partial \bar{x}^2}, \tag{7}$$

where the single remaining parameter is the **Reynolds number**

$$\boxed{\mathrm{Re} := \frac{UL}{\nu}} \tag{8}$$

which measures the strength of the nonlinear (advective) term relative to the linear (diffusive) one. From here on we drop the bars and work in dimensionless variables, i.e. $U = L = 1$ and $\nu = 1/\mathrm{Re}$. Reference [1] uses $L = 1$, $U = 1$, $\nu = 0.01$, hence $\mathrm{Re} = 100$.

Remark 2.1 (Why Re is the parameter that matters). Everything difficult about (1) is controlled by Re. At small Re diffusion dominates, the solution stays smooth and decays, and (as we show in Section 4.3) the Carleman embedding converges. At large Re the nonlinearity steepens the profile into a thin internal layer of width $\sim \mathrm{Re}^{-1}$, which must be resolved by the grid *and* which drives the Carleman series out of its radius of convergence. Nearly every cost in this report is really a cost in Re.

2.3 Nomenclature

Table 1 lists every symbol used in this report, in order of first appearance. Readers may prefer to skip it now and consult it as needed.

3 The pathway, stage by stage

This section restates the method of [1] with every step made explicit, because the resource analysis in Sections 4 and 5 refers to the individual stages. A reader already familiar with the paper can skim to Section 3.7, where we reproduce its published instance numerically.

3.1 Stage 1: spatial discretisation

Place n_x interior grid points on $[0, L]$,

$$x_j = j\,\Delta x, \quad j = 1, \ldots, n_x, \qquad \Delta x := \frac{L}{n_x + 1}, \tag{9}$$

with $x_0 = 0$ and $x_{n_x+1} = L$ carrying the boundary values. Write $u_j(t) \approx u(t, x_j)$ and collect them in $\boldsymbol{u}(t) = (u_1, \ldots, u_{n_x})^\top \in \mathbb{R}^{n_x}$. Replace the derivatives by second-order central differences (derived in Appendix A),

$$\left.\frac{\partial u}{\partial x}\right|_{x_j} \approx \frac{u_{j+1} - u_{j-1}}{2\Delta x}, \qquad \left.\frac{\partial^2 u}{\partial x^2}\right|_{x_j} \approx \frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2}, \tag{10}$$

each accurate to $\mathcal{O}(\Delta x^2)$. Substituting into (1) gives the semi-discrete system used in [1, Eq. (F.2)],

$$\frac{\mathrm{d}u_j}{\mathrm{d}t} = \underbrace{\frac{\nu}{\Delta x^2}\left(u_{j+1} - 2u_j + u_{j-1}\right)}_{\text{diffusion}} - \underbrace{\frac{1}{2\Delta x}u_j\left(u_{j+1} - u_{j-1}\right)}_{\text{advection}}, \qquad j = 1, \ldots, n_x. \tag{11}$$

Table 1: Nomenclature. "§" gives the section where the symbol is defined.

Symbol	Meaning	§
Continuum problem		
$u(t,x)$	scalar field solving Burgers' equation	2.1
ν	kinematic viscosity	2.1
$L,\ T$	domain length, final time	2.1
U	amplitude $\|u_0\|_\infty$ of the initial data	2.1
Re	Reynolds number UL/ν	2.2
ϵ	target relative ℓ^2 error at time T	2.1
δ_v	viscous-layer thickness, $\delta_\mathrm{v} \sim L/\mathrm{Re}$	4.2
Discretisation		
n_x	number of interior spatial grid points	3.1
Δx	grid spacing $L/(n_x+1)$	3.1
q	spatial qubits, $q = \lceil \log_2 n_x \rceil$	3.1
$\boldsymbol{u}(t)$	vector $(u_1,\ldots,u_{n_x})^\top$ of nodal values	3.1
$\mathsf{F}_1, \mathsf{F}_2$	linear and quadratic coefficient matrices	3.1
λ_1, λ_2	$\nu/\Delta x^2$ and $-1/(2\Delta x)$	3.1
Δt	time step	3.3
m	number of time steps, $m = T/\Delta t$	3.3
λ_diff	diffusion number $\nu\Delta t/\Delta x^2$	3.3
λ_cfl	Courant number $\Delta t/\Delta x$	3.3
Carleman embedding		
N	Carleman truncation order	3.2
$\hat{y}_j$	j-th Carleman block, $\boldsymbol{u}^{\otimes j}$	3.2
$\boldsymbol{y}$	stacked Carleman vector $(\hat{y}_1;\ldots;\hat{y}_N)$	3.2
A	Carleman matrix	3.2
A_k^j	block of A mapping $\hat{y}_k \to \dot{\hat{y}}_j$	3.2
D	dimension of A, $D = \mathcal{O}(n_x^N)$	3.2
$R,\ R_\star$	Carleman convergence ratios	4.3
γ	$\|\boldsymbol{u}\|_2$, controls block-1 readout	5.6
P_1	probability of the first Carleman block	5.6
Linear system and quantum algorithm		
L	implicit-step matrix $\mathbb{I} - \Delta t\,\mathsf{A}$	3.3
$\mathcal{L}$	dilated (history-state) matrix	5.7
α	block-encoding subnormalisation	3.4
s	maximum number of nonzeros per row of L	3.4
n_data	number of distinct (value, offset) pairs	5.3
m_flag	flag qubits (data + delete)	5.1
q_M	matrix-register qubits, $q_M = \lceil \log_2 D \rceil$	5.1
σ_k	singular values of $\mathsf{L}^\dagger/\alpha$	3.5
κ_Q	effective condition number $\alpha/\sigma_\mathrm{min}(\mathsf{L})$	5.2
κ_V	eigenvector condition number of A	5.7
d	degree of the QSVT inverse polynomial	3.5
β	QSP normalisation of the $1/x$ polynomial	3.5
p_s	QSVT post-selection success probability	3.5
p_tot	total success probability	5.6
$D_\mathrm{BE}, D_\Pi, D_\mathrm{QSVT}$	two-qubit depths	5.5
M_shot	number of measurement repetitions	5.8

This is a system of n_x coupled ordinary differential equations (ODEs) whose right-hand side is a *quadratic polynomial* in the unknowns. Writing it in the standard quadratic form

$$\boxed{\frac{\mathrm{d}\boldsymbol{u}}{\mathrm{d}t} = \mathsf{F}_1\,\boldsymbol{u} + \mathsf{F}_2\,\boldsymbol{u}^{\otimes 2}} \tag{12}$$

(the notation $\boldsymbol{u}^{\otimes 2} := \boldsymbol{u} \otimes \boldsymbol{u} \in \mathbb{R}^{n_x^2}$, the vector of all pairwise products $u_i u_j$, is explained in Appendix B) we read off, with

$$\lambda_1 := \frac{\nu}{\Delta x^2}, \qquad \lambda_2 := -\frac{1}{2\Delta x}, \tag{13}$$

the coefficient matrices [1, Eqs. (F.3)–(F.4)]

$$\mathsf{F}_1 = \lambda_1 \begin{pmatrix} -2 & 1 & & \\ 1 & -2 & 1 & \\ & \ddots & \ddots & \ddots \\ & & 1 & -2 \end{pmatrix} \in \mathbb{R}^{n_x \times n_x}, \qquad [\mathsf{F}_2]_{i,\, i\, n_x + j} = \begin{cases} +\lambda_2, & j = i+1, \\ -\lambda_2, & j = i-1, \\ 0, & \text{otherwise}, \end{cases} \tag{14}$$

with $\mathsf{F}_2 \in \mathbb{R}^{n_x \times n_x^2}$. The index pattern of F_2 says exactly what (11) says: the i-th output picks out the products $u_i u_{i+1}$ and $u_i u_{i-1}$.

Remark 3.1 (Conservative versus advective form)*.* Equation (11) discretises $u\,\partial_x u$ directly ("advective form"). The algebraically equivalent "conservative form" $\frac{1}{2}\partial_x(u^2)$ discretises to $-\frac{1}{4\Delta x}(u_{j+1}^2 - u_{j-1}^2)$ and has the advantage of exactly preserving a discrete analogue of the energy bound $\|u(t)\|_2 \le \|u_0\|_2$. Since the challenge specification supplies exactly that bound as the property that makes efficient simulation possible, the conservative (or the skew-symmetric average of the two) form is the better choice in practice. It changes the entries of F_2 but not its sparsity pattern, so all scaling results below are unaffected; we keep the advective form to match [1].

3.2 Stage 2: Carleman linearisation

The idea. Equation (12) is nonlinear only because of the term $\mathsf{F}_2\boldsymbol{u}^{\otimes 2}$. Carleman's observation is that if we *treat the products as new unknowns*, the equations governing them are again polynomial, and the whole system becomes linear — at the price of infinitely many variables. Define the Carleman blocks

$$\hat{y}_j(t) := \boldsymbol{u}(t)^{\otimes j} \in \mathbb{R}^{n_x^j}, \qquad j = 1, 2, 3, \ldots \tag{15}$$

so $\hat{y}_1 = \boldsymbol{u}$, $\hat{y}_2$ holds all quadratic monomials $u_a u_b$, $\hat{y}_3$ all cubic monomials, and so on.

The hierarchy. Differentiating $\hat{y}_j$ with the product rule (Appendix B, Lemma B.1) gives

$$\frac{\mathrm{d}\hat{y}_j}{\mathrm{d}t} = \sum_{v=0}^{j-1} \boldsymbol{u}^{\otimes v} \otimes \dot{\boldsymbol{u}} \otimes \boldsymbol{u}^{\otimes(j-1-v)}, \tag{16}$$

and substituting $\dot{\boldsymbol{u}} = \mathsf{F}_1\boldsymbol{u} + \mathsf{F}_2\boldsymbol{u}^{\otimes 2}$ into each slot,

$$\boxed{\frac{\mathrm{d}\hat{y}_j}{\mathrm{d}t} = A_j^j\,\hat{y}_j + A_{j+1}^j\,\hat{y}_{j+1}} \tag{17}$$

with the blocks [1, Eqs. (E.4)–(E.5)]

$$A_j^j = \sum_{v=0}^{j-1} \mathbb{I}^{\otimes v} \otimes \mathsf{F}_1 \otimes \mathbb{I}^{\otimes(j-1-v)} \;\in \mathbb{R}^{n_x^j \times n_x^j}, \tag{18}$$

$$A_{j+1}^j = \sum_{v=0}^{j-1} \mathbb{I}^{\otimes v} \otimes \mathsf{F}_2 \otimes \mathbb{I}^{\otimes(j-1-v)} \;\in \mathbb{R}^{n_x^j \times n_x^{j+1}}, \tag{19}$$

where $\mathbb{I}$ is the $n_x \times n_x$ identity.

The structure of (17) is worth pausing on, because it is the single most important fact about the method. Substituting the *linear* part $\mathsf{F}_1 \boldsymbol{u}$ into a degree-j product leaves the degree at j: that term closes on $\hat{y}_j$ itself. Substituting the *quadratic* part $\mathsf{F}_2 \boldsymbol{u}^{\otimes 2}$ raises the degree from j to $j+1$: that term reaches up to $\hat{y}_{j+1}$. So each equation depends on the next one, forever: $\hat{y}_1$ needs $\hat{y}_2$, which needs $\hat{y}_3$, and so on. The quadratic nonlinearity is not a bound on the hierarchy — it is the reason the hierarchy never closes.

Truncation. We close the system by hand at order N, setting $\hat{y}_{N+1} := 0$. Stacking $\boldsymbol{y} := (\hat{y}_1; \hat{y}_2; \ldots; \hat{y}_N)$ we obtain the finite *linear* ODE system [1, Eq. (E.2)]

$$\boxed{\frac{\mathrm{d}\boldsymbol{y}}{\mathrm{d}t} = \mathsf{A}\, \boldsymbol{y}}, \qquad \mathsf{A} = \begin{pmatrix} A_1^1 & A_2^1 & & \\ & A_2^2 & A_3^2 & \\ & & \ddots & \ddots \\ & & & A_N^N \end{pmatrix}, \tag{20}$$

which is block upper-bidiagonal. (In the general formulation of [1, Eq. (E.3)] there is also a subdiagonal A_{j-1}^{j} generated by an inhomogeneous term $F_0(t)$; for Burgers $F_0 = 0$ and those blocks vanish, as the paper notes below its Eq. (F.5).) The dimension is

$$D := \dim \mathsf{A} = \sum_{j=1}^{N} n_x^j = \frac{n_x^{N+1} - n_x}{n_x - 1} = \mathcal{O}(n_x^N)\,. \tag{21}$$

Remark 3.2 (Two different "orders"). Novices routinely conflate two numbers. The *degree of the nonlinearity* in (12) is 2 and is fixed by the PDE. The *Carleman truncation order* N is a free parameter of the algorithm, chosen to hit an accuracy target; $N = 2$ is merely the smallest choice that retains any nonlinear feedback. Larger N means a more faithful approximation and an n_x times larger system per unit increase.

3.3 Stage 3: implicit time stepping

Equation (20) is linear, but A is stiff: its spectrum spans many orders of magnitude (Appendix C). An explicit scheme would require $\Delta t \le \Delta x^2/(2\nu)$, which becomes prohibitive on fine grids. Reference [1] therefore uses backward (implicit) Euler. With time levels $t_k = k\Delta t$ and $\boldsymbol{y}^k \approx \boldsymbol{y}(t_k)$,

$$\frac{\boldsymbol{y}^{k+1} - \boldsymbol{y}^k}{\Delta t} = \mathsf{A}\, \boldsymbol{y}^{k+1} \quad \Longleftrightarrow \quad \boxed{\mathsf{L}\, \boldsymbol{y}^{k+1} = \boldsymbol{y}^k, \qquad \mathsf{L} := \mathbb{I}_D - \Delta t\, \mathsf{A}} \tag{22}$$

where $\mathbb{I}_D$ is the $D \times D$ identity. This is [1, Eq. (E.8)]. It is unconditionally stable, and each step is a linear system — which is precisely what the quantum subroutine solves. We will use two dimensionless step-size measures throughout,

$$\lambda_{\text{diff}} := \frac{\nu\, \Delta t}{\Delta x^2} = \frac{\Delta t}{\mathrm{Re}\, \Delta x^2} \quad \text{(diffusion number)}, \qquad \lambda_{\text{cfl}} := \frac{\Delta t}{\Delta x} \quad \text{(Courant number)}, \tag{23}$$

and $m := T/\Delta t$ for the number of steps.

3.4 Stage 4: block encoding

A quantum computer applies unitaries, and L is not unitary. A **block encoding** embeds it as the top-left corner of a larger unitary (Appendix E): a unitary U_{L} acting on $q_M + m_{\text{flag}}$ qubits is a block encoding of L with subnormalisation α if

$$\big(\langle 0^{m_{\text{flag}}}| \otimes \mathbb{I}\big)\, U_{\mathsf{L}}\, \big(|0^{m_{\text{flag}}}\rangle \otimes \mathbb{I}\big) = \frac{\mathsf{L}}{\alpha}, \qquad \alpha \ \ge\ \|\mathsf{L}\|_2, \tag{24}$$

i.e. $U_{\mathsf{L}} = \begin{pmatrix} \mathsf{L}/\alpha & * \\ * & * \end{pmatrix}$ in the basis where the first block corresponds to all flag qubits being $|0\rangle$. The division by α is forced: a unitary has all singular values equal to 1, so any submatrix has singular values in $[0, 1]$ and L must be scaled down to fit. *The value of α is not a bookkeeping detail — it directly multiplies the cost*, as Section 5.2 shows.

3.5 Stage 5: QSVT matrix inversion

Let the singular value decomposition of the subnormalised, conjugate-transposed matrix be

$$\frac{\mathsf{L}^\dagger}{\alpha} = \sum_k \sigma_k \,|v_k\rangle\langle w_k|, \qquad \sigma_k \in [0, 1], \tag{25}$$

with $\{|v_k\rangle\}$, $\{|w_k\rangle\}$ orthonormal. Solving $\mathsf{L}\boldsymbol{y} = \boldsymbol{b}$ means applying $\sum_k \sigma_k^{-1}|w_k\rangle\langle v_k|$, i.e. applying the function $f(\sigma) = 1/\sigma$ to every singular value. The quantum singular value transformation (QSVT, Appendix D) does exactly this for any polynomial Poly of degree d that is bounded by 1 on $[-1, 1]$: interleaving d copies of U_{L} and $U_{\mathsf{L}}^\dagger$ with $d+1$ projector-controlled phase rotations Π_{ϕ_i}, $i = 0, \ldots, d$ (multi-controlled rotations that fire only when the flag register is $|0\rangle$; see Appendix D) produces a circuit $U_{\vec{\phi}}$ whose top-left block is $\sum_k \mathrm{Poly}(\sigma_k)|w_k\rangle\langle v_k|$ [1, Theorem 2].

Since $1/\sigma$ is unbounded at $\sigma = 0$ it cannot be approximated by a polynomial on all of $[0, 1]$. One therefore fixes a *design cut-off* $\hat{\sigma}_{\min} = 1/\kappa$ and approximates $1/\sigma$ only on $[1/\kappa, 1]$, using the Chebyshev construction of [1, Appendix B],

$$\mathrm{Poly}(\sigma) \;\approx\; \frac{\beta}{\sigma} \quad \text{on } \sigma \in [1/\kappa, 1], \qquad d \;=\; \mathcal{O}\big(\kappa \log(\kappa/\epsilon_{\mathrm{inv}})\big), \tag{26}$$

where $\beta \le 1/(2\kappa)$ is a normalisation forced by the requirement $|\mathrm{Poly}| \le 1$, and ϵ_{inv} is the accuracy of the polynomial approximation. Applying $U_{\vec{\phi}}$ to $|b\rangle$ and post-selecting all flag and QSP qubits on $|0\rangle$ yields the normalised solution $|\hat{y}\rangle$ with probability [1, Eq. (7)]

$$p_s = \sum_k \Big(\frac{\beta}{\sigma_k}\Big)^2 |\langle w_k|b\rangle|^2 = \beta^2 \left\|(\mathsf{L}^\dagger/\alpha)^{-1}|b\rangle\right\|^2. \tag{27}$$

3.6 Stage 6: readout

The circuit outputs $|\hat{y}(T)\rangle$, the normalised *Carleman* vector. Two further extractions are needed and neither is free:

1. the physical field lives in the first block $\hat{y}_1$, so one must post-select the block register on "block 1" (Section 5.6);
2. the amplitude encoding (4) carries no overall scale, so $\|\hat{\boldsymbol{u}}(T)\|_2$ must be estimated separately (Section 5.9).

Reference [1] lists three readout options — full tomography, sampling the distribution, or estimating observables $\langle x|\mathcal{M}|x\rangle$ — and notes correctly that the first sacrifices any exponential speedup.

3.7 The reference instance, reproduced

Section 4.3 of [1] reports a concrete Burgers instance. We reconstructed it from the equations above; Table 2 compares our reconstruction with the published values. Agreement is exact for every structural quantity and within 5% for the success probability, which validates both our reading of the method and the models developed in Section 5.

Two entries in Table 2 deserve immediate comment and are developed later. First, the design value $\kappa = 8$ chosen in [1] is close to our $\kappa_{\mathrm{Q}} = 5.46$ and far from the classical condition number 1.57; this is the evidence for finding (i) of Section 1.3. Second, the Carleman convergence ratio is $R = 76 \gg 1$, so the truncation at $N = 2$ has no convergence guarantee; this is finding (iv).

Table 2: Reconstruction of the Burgers instance of [1, §4.3, Tables 3–4]. Parameters: $S = 7$ total grid points so $n_x = S - 2 = 5$, $\Delta x = 1/(S-1) = 1/6$, $\nu = 0.01$, $\Delta t = 0.1$, $T = 0.3$, $N = 2$, $u_0(x) = \sin(2\pi x)$. Hence $\mathrm{Re} = 100$, $\lambda_1 = 0.36$, $\lambda_2 = -3$, $\lambda_{\mathrm{diff}} = 0.036$, $\lambda_{\mathrm{cfl}} = 0.6$.

Quantity	Formula	This work	[1]
System dimension D	$n_x + n_x^2$	30	30
Matrix qubits q_M	$\lceil \log_2 D \rceil$	5	5
Data elements n_{data}	Eq. (50)	14	14
Data qubits	$\lceil \log_2 n_{\mathrm{data}} \rceil$	4	4
Total qubits	$q_M + n_{\mathrm{data}}$ qubits $+1+1$	11	11
Distinct values	$\{\pm 0.3, -0.036, 1.072, 1.144\}$	identical to Table 3 of [1]	
Subnormalisation α	Eq. (49)	4.760	(not reported)
$\sigma_{\max}(\mathsf{L})$, $\sigma_{\min}(\mathsf{L})$	—	1.370, 0.872	(not reported)
$\sigma_{\max}/\sigma_{\min}$	classical condition number	1.571	—
$\kappa_{\mathrm{Q}} = \alpha/\sigma_{\min}$	Eq. (47)	**5.457**	$\kappa = 8$ (design)
Success probability p_s	Eq. (58)	**0.0911**	**0.0865**
Carleman ratio R	Eq. (33)	**76.2**	(not reported)
Block-1 probability P_1	Eq. (59)	0.250	(not discussed)

4 Error budget

Every stage of Section 3 introduces error. This section enumerates them, quantifies each, and — crucially — establishes which are amplified by the condition number and which are not, since that determines how tightly each must be controlled.

4.1 Taxonomy

Write $\boldsymbol{u}^{\mathrm{exact}}(T)$ for the exact PDE solution on the grid and $\hat{\boldsymbol{u}}(T)$ for what the algorithm returns. Inserting intermediate quantities and using the triangle inequality,

$$\underbrace{\|\hat{\boldsymbol{u}} - \boldsymbol{u}^{\mathrm{exact}}\|}_{\text{total}} \le \underbrace{\varepsilon_{\mathrm{sp}}}_{§4.2} + \underbrace{\varepsilon_{\mathrm{CL}}}_{§4.3} + \underbrace{\varepsilon_{\mathrm{t}}}_{§4.4} + \underbrace{\varepsilon_{\mathrm{alg}}}_{§4.5} + \underbrace{\varepsilon_{\mathrm{stat}}}_{§4.6}, \tag{28}$$

with the five contributions listed in Table 3. The last column is the key structural fact of this section.

4.2 Spatial discretisation error

The stencils (10) have local truncation errors (Taylor expansions in Appendix A)

$$\tau_j = \frac{\nu\,\Delta x^2}{12}\,\partial_x^4 u(x_j) - \frac{\Delta x^2}{6}\,u\,\partial_x^3 u(x_j) + \mathcal{O}(\Delta x^4). \tag{29}$$

Local errors accumulate. Writing $e(t) := \hat{\boldsymbol{u}}(t) - \boldsymbol{u}^{\mathrm{exact}}(t)$ and subtracting (11) from the exact equation, a Gronwall argument (Appendix A, Lemma A.1) gives

$$\|e(T)\| \;\le\; \frac{e^{\Lambda T} - 1}{\Lambda}\max_{t \le T}\|\tau(t)\|, \qquad \Lambda := \|\partial_x u\|_\infty \tag{30}$$

where Λ is the one-sided Lipschitz constant of the semi-discrete right-hand side; the advective term supplies $\|\partial_x u\|_\infty$ and the diffusive term only helps. Hence

$$\varepsilon_{\mathrm{sp}} \;\simeq\; C_{\mathrm{sp}}\,\Delta x^2, \qquad C_{\mathrm{sp}} = \frac{e^{\Lambda T} - 1}{\Lambda}\Big(\tfrac{\nu}{12}\|\partial_x^4 u\| + \tfrac{1}{6}\|u\,\partial_x^3 u\|\Big). \tag{31}$$

Table 3: The error sources of the pathway. "$\times\kappa_Q$" marks errors that are amplified by the effective condition number before reaching the output (Section 4.5 and Appendix G). Symbols used here and defined in the sections indicated: $\Lambda = \|\partial_x u\|_\infty$ is the velocity-gradient rate of Eq. (30); R is the Carleman ratio of Eq. (33); M_{shot} is the number of circuit repetitions (§5.9); D_{QSVT} is the circuit depth (§5.5); and ε_{2q} is the two-qubit gate infidelity of the hardware.

	Source	Magnitude	Amplified?
ε_{sp}	finite-difference truncation, Eq. (10)	$\mathcal{O}(\Delta x^2)e^{\Lambda T}$	no
ε_{CL}	Carleman truncation at order N	$\mathcal{O}(R^N)$	no
ε_{t}	backward Euler, Eq. (22)	$\mathcal{O}(\Delta t\, T\|\mathsf{A}\|^2)$	no
ε_b	error of the input oracle (3)	ϵ_b	$\times\kappa_Q$
ε_{BE}	finite precision of the block-encoding data	ϵ_{BE}	$\times\kappa_Q$
ε_{inv}	polynomial approximation of $1/\sigma$	ϵ_{inv}	$\times\kappa_Q$
ε_ϕ	classical computation of the phases $\vec{\phi}$	ϵ_ϕ	$\times\kappa_Q$
ε_{cut}	singular values below the cut-off $1/\kappa$	see §4.5	hard
$\varepsilon_{\text{stat}}$	shot noise / tomography	$\mathcal{O}(M_{\text{shot}}^{-1/2})$	no
$\varepsilon_{\text{norm}}$	estimation of $\|\hat{\boldsymbol{u}}(T)\|_2$	$\mathcal{O}(M_{\text{shot}}^{-1/2})$	no
ε_{hw}	hardware noise (not fault-tolerant)	$\mathcal{O}(D_{\text{QSVT}}\,\varepsilon_{2q})$	no

Two separate resolution requirements. Setting $\varepsilon_{\text{sp}} = \epsilon$ gives the accuracy requirement $n_x \gtrsim \sqrt{C_{\text{sp}}}\,\epsilon^{-1/2}$. But there is a second, harder constraint. At Reynolds number Re the Burgers solution develops an internal layer of thickness $\delta_{\text{v}} \sim L/\text{Re}$, and central differences produce spurious oscillations unless the *cell Reynolds number* satisfies $\text{Re}_\Delta := U\Delta x/\nu \lesssim 2$. Combining,

$$\boxed{n_x \;\gtrsim\; \max\!\left(\tfrac{1}{2}\text{Re},\;\sqrt{C_{\text{sp}}}\,\epsilon^{-1/2}\right).} \tag{32}$$

For the published instance, $\text{Re} = 100$ requires $n_x \gtrsim 50$, whereas [1] uses $n_x = 5$: the demonstration is under-resolved by an order of magnitude in each direction. Again, this is appropriate for a circuit demonstration and merely means the reported figure 5(b) should not be read as a converged solution.

4.3 Carleman truncation error

This is the error introduced by setting $\hat{y}_{N+1} = 0$ in (17), and it is the deepest limitation of the whole pathway.

The convergence criterion. The standard analysis [9, 10] of the quadratic system (12) requires F_1 to be dissipative — all eigenvalues with negative real part — and defines the dimensionless ratio

$$\boxed{R := \frac{\|\boldsymbol{u}_0\|_2\,\|\mathsf{F}_2\|_2}{|\text{Re}\,\mu_1(\mathsf{F}_1)|}}, \qquad \mu_1(\mathsf{F}_1) := \text{eigenvalue of } \mathsf{F}_1 \text{ of largest real part}, \tag{33}$$

which compares the rate at which the nonlinearity feeds energy up the hierarchy against the rate at which diffusion removes it. If $R < 1$ the truncation error obeys, uniformly in t,

$$\varepsilon_{\text{CL}} \;\lesssim\; \|\boldsymbol{u}_0\|_2\,R^N \qquad\Longrightarrow\qquad \boxed{N = \left\lceil \frac{\ln(1/\epsilon_{\text{CL}})}{\ln(1/R)} \right\rceil}. \tag{34}$$

If $R \geq 1$ no such bound exists: increasing N is not guaranteed to help.

Evaluating R for this discretisation. Both ingredients can be computed in closed form (Appendix C). The eigenvalues of F_1 in (14) are

$$\mu_k(\mathsf{F}_1) = -\frac{4\nu}{\Delta x^2}\sin^2\!\Big(\frac{k\pi}{2(n_x+1)}\Big), \qquad k = 1,\dots,n_x, \tag{35}$$

all real and negative, so F_1 is dissipative as required. The one closest to zero is $k = 1$; for $n_x \gg 1$, expanding $\sin\theta \approx \theta$,

$$|\mathrm{Re}\,\mu_1| = \frac{4\nu}{\Delta x^2}\sin^2\!\Big(\frac{\pi\Delta x}{2L}\Big) \xrightarrow{\Delta x\to 0} \frac{\pi^2\nu}{L^2}. \tag{36}$$

(Numerically, at $n_x = 5$ this predicts 0.0987 against an exact 0.0965, a 2% error.) For the nonlinear part, each row of F_2 has two entries of magnitude $1/(2\Delta x)$ and each column at most one, so $\|\mathsf{F}_2\|_2 \le \sqrt{\|\mathsf{F}_2\|_1\|\mathsf{F}_2\|_\infty} = 1/(\sqrt{2}\,\Delta x)$. With $u_0 = U\sin(2\pi x/L)$ we have $\|\boldsymbol{u}_0\|_2 = U\sqrt{n_x/2}$. Hence

$$R \;\simeq\; \frac{U\sqrt{n_x/2}\cdot\big(\sqrt{2}\,\Delta x\big)^{-1}}{\pi^2\nu/L^2} = \frac{UL^2\sqrt{n_x}}{2\pi^2\nu\,\Delta x} \;\simeq\; \frac{\mathrm{Re}\;n_x^{3/2}}{2\pi^2}. \tag{37}$$

The grid-dependence is an artefact, but the conclusion is not. Equation (37) grows like $n_x^{3/2}$, which cannot be physically meaningful: refining the grid does not make the PDE more nonlinear. The inflation comes from using the unweighted ℓ^2 operator norm of F_2, which measures its action on *all* of $\mathbb{R}^{n_x^2}$, whereas the dynamics only ever visits the tensor-product manifold $\{\boldsymbol{u}\otimes\boldsymbol{u}\}$. On that manifold,

$$\big[\mathsf{F}_2\,\boldsymbol{u}^{\otimes 2}\big]_j = -\frac{u_j(u_{j+1}-u_{j-1})}{2\Delta x} \;\approx\; -u_j\,(\partial_x u)_j \quad\Longrightarrow\quad \frac{\|\mathsf{F}_2\boldsymbol{u}^{\otimes 2}\|}{\|\boldsymbol{u}\|} \le \|\partial_x u\|_\infty, \tag{38}$$

which is grid-independent. Substituting this physical rate and (36) into (33), and using $\|\partial_x u\|_\infty = 2\pi U/L$ for $u_0 = U\sin(2\pi x/L)$, gives the sharpened ratio

$$\boxed{R_\star = \frac{\|\partial_x u\|_\infty}{\pi^2\nu/L^2} = \frac{2\pi U/L}{\pi^2\nu/L^2} = \frac{2}{\pi}\,\mathrm{Re}} \qquad\Longrightarrow\qquad R_\star < 1 \iff \mathrm{Re} < \frac{\pi}{2} \approx 1.57. \tag{39}$$

This is the central negative result about the Carleman stage, and it is robust: the naive estimate (37) gives $R = 56.6$ for the published instance, the sharpened estimate (39) gives $R_\star = 63.7$, and evaluating (33) directly from the assembled matrices gives $R = 76.2$ (Appendix I). All three agree that $\mathrm{Re} = 100$ is two orders of magnitude outside the convergent regime. Figure 1(a) shows the required order N from (34) as a function of Re: it diverges as $\mathrm{Re} \to \pi/2$ and does not exist beyond.

Remark 4.1 (Rescaling does not help). A natural thought is to rescale $\boldsymbol{u} \to \boldsymbol{u}/\Gamma$ to shrink $\|\boldsymbol{u}_0\|$. But (12) is then satisfied with $\mathsf{F}_2 \to \Gamma\mathsf{F}_2$, so the product $\|\boldsymbol{u}_0\|\|\mathsf{F}_2\|$ in (33) is unchanged: R is invariant under rescaling. This is worth internalising — the constraint (39) is physical, not a normalisation artefact. (Rescaling *does* help with a different problem, the block-1 readout of Section 5.6.)

Remark 4.2 (Short-time bounds). For non-dissipative problems (e.g. the Korteweg–de Vries equation, where F_1 is skew-symmetric and $\mathrm{Re}\,\mu_1 = 0$ exactly, so $R = \infty$), criterion (33) is vacuous and one must use finite-horizon bounds of the form $\varepsilon_{\mathrm{CL}} \lesssim \|\boldsymbol{u}_0\|(T/\tau_{\mathrm{nl}})^N$ with $\tau_{\mathrm{nl}} = \|\partial_x u\|_\infty^{-1}$ the nonlinear time scale [11]. These converge only for $T < \tau_{\mathrm{nl}}$, roughly one eddy turnover. Restarting is not an option, because restarting requires re-preparing an amplitude encoding of $\hat{\boldsymbol{u}}(T)$, which requires tomography.

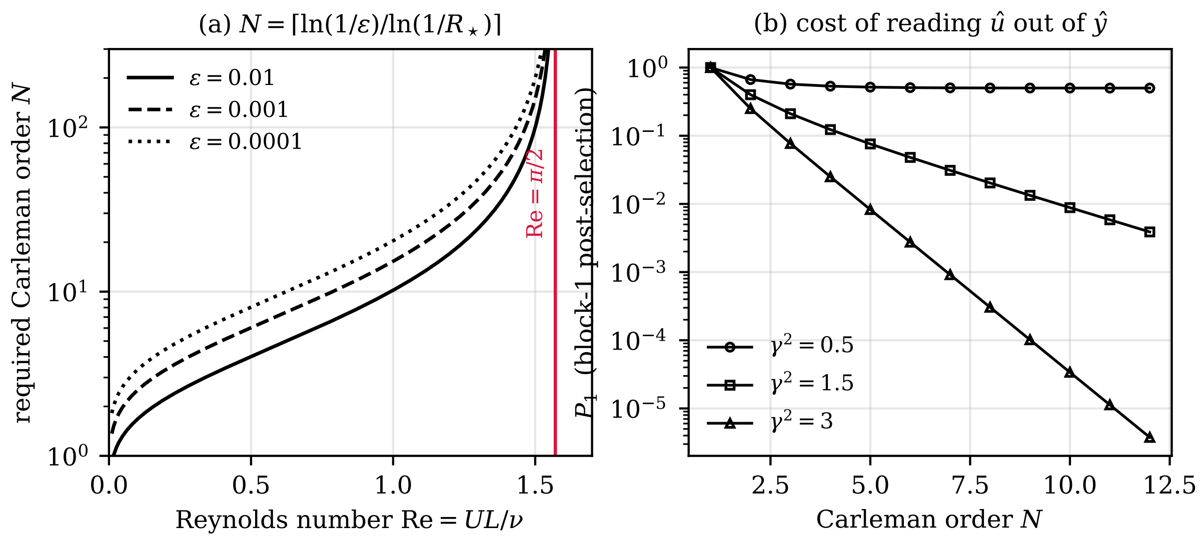

Figure 1: (a) Carleman order N required by (34) to reach relative error ϵ, using the sharpened ratio $R_\star = 2\mathrm{Re}/\pi$ of (39). Beyond $\mathrm{Re} = \pi/2$ (red line) the series has no proven convergence regime and no finite N suffices. (b) The block-1 post-selection probability P_1 of Eq. (59), which decays geometrically in N unless the state is rescaled so that $\gamma = \|\boldsymbol{u}\|_2 < 1$.

4.4 Time discretisation error

Backward Euler is first-order accurate. For the linear system $\dot{\boldsymbol{y}} = \mathsf{A}\boldsymbol{y}$ the global error after time T is (Appendix A)

$$\varepsilon_{\mathrm{t}} \;\simeq\; \tfrac{1}{2}\,\Delta t\, T\,\|\mathsf{A}\|^2\|\boldsymbol{y}\|, \qquad \Longrightarrow \qquad \Delta t = \frac{2\,\epsilon_{\mathrm{t}}}{T\|\mathsf{A}\|^2}, \quad m = \frac{T}{\Delta t} = \frac{T^2\|\mathsf{A}\|^2}{2\,\epsilon_{\mathrm{t}}}. \tag{40}$$

From (18)–(19), each block is a sum of $j \le N$ terms, so

$$\|\mathsf{A}\| \;\le\; N \max\bigl(\|\mathsf{F}_1\|, \|\mathsf{F}_2\|\|\boldsymbol{u}\|\bigr) = N\,\Lambda_{\mathsf{A}}, \qquad \Lambda_{\mathsf{A}} := \max\Bigl(\frac{4\nu}{\Delta x^2}, \frac{U}{\sqrt{2}\,\Delta x}\Bigr). \tag{41}$$

Whenever the grid resolves the viscous layer, $\Delta x \lesssim \delta_{\mathrm{v}} = \nu/U$, the diffusive entry dominates and $\Lambda_{\mathsf{A}} \simeq 4\nu/\Delta x^2 = 4n_x^2/(\mathrm{Re}L^2)$. Substituting into (40) gives $m = \mathcal{O}(N^2 n_x^4/(\mathrm{Re}^2\epsilon_{\mathrm{t}}))$, i.e. $m = \mathcal{O}(\epsilon^{-3})$ once $n_x \propto \epsilon^{-1/2}$. This is unacceptably steep and is a defect of first-order stepping, not of the quantum algorithm. Replacing backward Euler by a high-order scheme — the truncated Taylor or linear-multistep encodings of [7, 8] — gives instead

$$\boxed{m = \mathcal{O}\Bigl(\|\mathsf{A}\|\,T\,\frac{\log(1/\epsilon_{\mathrm{t}})}{\log\log(1/\epsilon_{\mathrm{t}})}\Bigr) = \widetilde{\mathcal{O}}\Bigl(\frac{N\,n_x^2\,T}{\mathrm{Re}}\Bigr),} \tag{42}$$

which we adopt for all asymptotic statements below. We flag this as a recommended modification to the pathway of [1]: the change is purely classical (a different sparse matrix to block-encode) and costs nothing in qubits.

4.5 Algorithmic errors and why they are amplified

The errors in the middle group of Table 3 enter differently from the discretisation errors, because they perturb the linear system rather than the model. If the block encoding realises $\mathsf{L} + \delta\mathsf{L}$ instead of L, and the oracle prepares $|b\rangle + |\delta b\rangle$, then standard perturbation theory (Appendix G, Lemma G.1) gives

$$\frac{\|\delta\boldsymbol{y}\|}{\|\boldsymbol{y}\|} \;\le\; \frac{\kappa_{\mathrm{Q}}}{1 - \kappa_{\mathrm{Q}}\|\delta\mathsf{L}\|/\|\mathsf{L}\|}\left(\frac{\|\delta\mathsf{L}\|}{\|\mathsf{L}\|} + \frac{\|\delta b\|}{\|b\|}\right) \;\approx\; \kappa_{\mathrm{Q}}\,(\epsilon_{\mathrm{BE}} + \epsilon_b)\,, \tag{43}$$

and the same factor multiplies the polynomial-approximation error ϵ_{inv} and the phase-synthesis error ϵ_ϕ. The practical consequence is that these four quantities must be controlled to $\epsilon/\kappa_{\mathrm{Q}}$ rather than ϵ, which adds a $\log \kappa_{\mathrm{Q}}$ to the polynomial degree (26) and to the arithmetic precision of the block encoding.

One error in this group behaves qualitatively differently. The polynomial $\mathrm{Poly}(\sigma)$ approximates $1/\sigma$ only on $[1/\kappa, 1]$; any genuine singular value of $\mathsf{L}^\dagger/\alpha$ *below* the design cut-off is not inverted but suppressed. This is not a small perturbation — it deletes a spectral component — so κ must be chosen with $1/\kappa \le \sigma_{\min}$ guaranteed, not merely likely. Reference [1] handles this correctly by estimating $\sigma_{\min}$ classically first (its §2), and by choosing κ conservatively; note this is the reason its $\kappa = 8$ exceeds our computed $\kappa_{\mathrm{Q}} = 5.46$.

4.6 Statistical error

Finally, the output is a quantum state, and every number extracted from it carries shot noise $\mathcal{O}(M_{\mathrm{shot}}^{-1/2})$ where M_{shot} is the number of repetitions. How many repetitions are needed depends entirely on what is being asked for, and is quantified in Section 5.9.

4.7 Budget allocation

We allocate the total budget (5) equally across the five groups, $\varepsilon_\bullet = \epsilon/5$, which is optimal up to constants because each cost depends only logarithmically or polynomially on its own tolerance. This fixes

$$n_x \gtrsim \max\left(\tfrac{1}{2}\mathrm{Re},\ \sqrt{5C_{\mathrm{sp}}/\epsilon}\right), \quad N = \left\lceil \frac{\ln(5/\epsilon)}{\ln(\pi/2\mathrm{Re})} \right\rceil, \quad m = \widetilde{\mathcal{O}}\left(\frac{N n_x^2 T}{\mathrm{Re}}\right), \quad \epsilon_{\mathrm{inv}} = \epsilon_{\mathrm{BE}} = \frac{\epsilon}{5\kappa_{\mathrm{Q}}}. \tag{44}$$

5 Resource estimates

5.1 Qubit count

The register structure of [1, Fig. 2] is: one QSP qubit carrying the phase sequence; a flag register of m_{flag} qubits (the data qubits of the block encoding plus one "delete" qubit); and a matrix register of q_M qubits holding $|b\rangle$ and the encoded matrix. Hence

$$\boxed{\#\text{qubits} = \underbrace{q_M}_{\lceil \log_2 D \rceil} + \underbrace{m_{\mathrm{flag}}}_{\lceil \log_2 n_{\mathrm{data}} \rceil + 1} + \underbrace{1}_{\mathrm{QSP}} + \underbrace{\mathcal{O}(1)}_{\text{amplitude amplification}}} \tag{45}$$

with, from (21) and writing $q = \log_2 n_x$,

$$q_M = \lceil \log_2 D \rceil \simeq Nq = N \log_2 n_x. \tag{46}$$

For the reference instance, $q_M = \lceil \log_2 30 \rceil = 5$, $m_{\mathrm{flag}} = \lceil \log_2 14 \rceil + 1 = 5$, total $5 + 5 + 1 = 11$, matching [1, Table 4] exactly.

Equation (46) is the pathway's one unambiguous success: the state register is *logarithmic* in the grid size and *linear* in the Carleman order, so a 2^{10}-point grid at $N = 6$ needs only 60 matrix qubits where a classical solver would store 10^{18} numbers. Everything that follows is the story of how that compression is paid for elsewhere.

5.2 Subnormalisation and the effective condition number

Which condition number? Reference [1] writes κ for "the condition number" without specifying whether it refers to L or to the subnormalised L/α. The distinction matters, because the polynomial in (26) must invert the singular values of $\mathsf{L}^\dagger/\alpha$, which are $\sigma_k(\mathsf{L})/\alpha$. The cut-off must therefore satisfy $1/\kappa \le \sigma_{\min}(\mathsf{L})/\alpha$, i.e.

$$\boxed{\kappa_{\mathrm{Q}} \;:=\; \frac{\alpha}{\sigma_{\min}(\mathsf{L})}} \qquad \text{not} \qquad \frac{\sigma_{\max}(\mathsf{L})}{\sigma_{\min}(\mathsf{L})}. \tag{47}$$

The two coincide only if α is chosen equal to $\sigma_{\max}$, which is not achievable for a sparse-oracle block encoding. Table 4 settles the question empirically against the three worked examples of [1]: κ_{Q} tracks the design values, the classical condition number does not.

Table 4: Which condition number sets the QSVT degree? Reconstructed from the three examples of [1]. The design value κ is what the paper used to synthesise the phases. $\kappa_{\mathrm{Q}} = \alpha/\sigma_{\min}$ tracks it to within the deliberate safety margin; $\sigma_{\max}/\sigma_{\min}$ does not.

Instance	α	$\sigma_{\max}$	$\sigma_{\min}$	$\dfrac{\sigma_{\max}}{\sigma_{\min}}$	$\kappa_{\mathrm{Q}} = \dfrac{\alpha}{\sigma_{\min}}$	design κ of [1]
Burgers, §4.3 ($n_x = 5$, $N = 2$)	4.76	1.37	0.872	1.57	5.46	8
Heat, §4.2 ($q_M = 3$)	4.84	3.57	1.014	3.52	4.77	8
Heat, §4.2 ($q_M = 4$)	4.84	3.57	1.000	3.57	4.84	12

Computing α in closed form. For the coherent-permutation block encoding of [2] used in [1], the subnormalisation is the ℓ^1 norm of the *data vector* — the list of distinct matrix values, each tagged with the diagonal offset (row minus column) on which it repeats:

$$\alpha = \|v_{\mathrm{data}}\|_1 = \sum_i |\psi_i|, \qquad n_{\mathrm{data}} = \#\{(\text{value},\ \text{offset})\ \text{pairs of}\ \mathsf{L}\}. \tag{48}$$

Counting these pairs for $\mathsf{L} = \mathbb{I} - \Delta t \mathsf{A}$ using (18)–(19):

- *Diagonals.* Block j has diagonal entry $1+2j\lambda_{\mathrm{diff}}$ (each of the j copies of F_1 contributes $-2\lambda_1\Delta t$), giving N distinct values summing to $N + N(N+1)\lambda_{\mathrm{diff}}$.
- *Off-diagonal* F_1 *entries.* In block j the copies of F_1 sit at offsets $\pm n_x^v$, $v = 0, \ldots, j-1$. Across all blocks these coincide, leaving $2N$ distinct offsets, each of magnitude λ_{diff}.
- F_2 *entries.* These live in the *rectangular* blocks A^j_{j+1}, whose entries do *not* lie on constant diagonals of the square embedding: the offset varies from row to row. This produces $\Theta(n_x^{N-1})$ distinct pairs, each of magnitude $\lambda_{\mathrm{cfl}}/2$.

Hence

$$\boxed{n_{\mathrm{data}} = N + 2N + \Theta\big(n_x^{N-1}\big)\,, \qquad \alpha = \underbrace{N + N(N+1)\lambda_{\mathrm{diff}} + 2N\lambda_{\mathrm{diff}}}_{\text{linear part}} + \underbrace{\Theta\big(n_x^{N-1}\big)\,\frac{\lambda_{\mathrm{cfl}}}{2}}_{\text{nonlinear part}}.} \tag{49}$$

For $N = 2$ the last count is exactly $2(n_x - 1)$, giving $n_{\mathrm{data}} = 3N + 2(n_x - 1)$, which for $n_x = 5$, $N = 2$ evaluates to 14 — the number of data elements in [1, Table 3]. Equation (50) below records the exact $N = 2$ formula. The α formula is verified to be *exact* against direct construction of the matrices over a range of Δt spanning three orders of magnitude (Appendix I).

$$n_{\mathrm{data}}\big|_{N=2} = 3N + 2(n_x - 1) = 6 + 2(n_x - 1). \tag{50}$$

The bad news. Because α enters κ_{Q} and hence the polynomial degree linearly, (49) says

$$\kappa_{\mathrm{Q}} \;\simeq\; \alpha \;\simeq\; n_x^{N-1}\lambda_{\mathrm{cfl}} \qquad \text{(coherent-permutation block encoding)}, \tag{51}$$

where we used $\sigma_{\min}(\mathsf{L}) = \mathcal{O}(1)$, which holds because A has eigenvalues with non-positive real part so $\mathsf{L} = \mathbb{I} - \Delta t \mathsf{A}$ has eigenvalues of modulus ≥ 1. This is polynomial in the grid size and *exponential in the Carleman order* — precisely the scaling that the logarithmic qubit count (46) was supposed to avoid. Figure 2 shows the measured growth.

5.3 Two block encodings, and why the choice dominates everything

The scaling (51) is a property of the *construction*, not of block encoding as such. The coherent-permutation scheme of [2] loads all distinct matrix values into a data register and permutes them onto constant diagonals; it is excellent for Toeplitz-like matrices from finite differences, which is what it was designed for, and poorly matched to the rectangular Kronecker blocks A^j_{j+1} of the Carleman matrix.

The standard alternative is the *sparse-access oracle* block encoding (Appendix E). It requires two oracles: O_{loc}, which given a row index and a counter $\ell \le s$ returns the column index of the ℓ-th nonzero; and O_{L}, which returns the corresponding value. Both are computed *arithmetically* from the closed-form structure (18)–(19) rather than looked up in a table, and the subnormalisation is

$$\alpha_{\text{sp}} = s\,\|\mathsf{L}\|_{\max}, \qquad s = \max_{\text{rows}} \#\{\text{nonzeros}\} = 4N-3, \qquad \|\mathsf{L}\|_{\max} = 1 + 2N\lambda_{\text{diff}}, \tag{52}$$

so that

$$\boxed{\alpha_{\text{sp}} = (4N-3)(1+2N\lambda_{\text{diff}}) = \mathcal{O}\big(N^2\lambda_{\text{diff}}\big)\,,} \tag{53}$$

independent of n_x except through the diffusion number. The row sparsity $s = 4N-3$ is verified numerically (Appendix I): each row of A^j_j has $2j+1$ nonzeros and each row of A^j_{j+1} has $2j$, but the maxima are not attained simultaneously. Table 5 quantifies the gap.

Table 5: Subnormalisation for the two block-encoding routes ($\nu = 0.01$, $\Delta t = 0.1$). α_{cp} is the coherent-permutation value of [1], computed exactly from the assembled matrix; $\alpha_{\text{sp}} = s\|\mathsf{L}\|_{\max}$ is the sparse-oracle value. The final column is the factor by which the published construction inflates the QSVT degree.

n_x	N	D	n_{data}	α_{cp}	α_{sp}	ratio
5	2	30	14	4.76	5.72	0.83
7	2	56	18	7.44	6.28	1.18
9	2	90	22	11.00	7.00	1.57
15	2	240	34	26.96	10.12	2.66
21	2	462	46	50.84	13.24	3.84
5	3	155	65	20.45	10.94	1.87
7	3	399	117	47.35	12.46	3.80
9	3	819	185	92.80	14.40	6.44
15	3	3615	485	388.41	22.82	17.02

The trade. The sparse oracle buys a much smaller α at the cost of quantum arithmetic: O_{loc} must add and compare q_M-bit integers, which needs ripple-carry adders of depth $\mathcal{O}(q_M)$ and $\mathcal{O}(q_M)$ ancillas, and O_{L} must synthesise a rotation angle from the entry value. The coherent-permutation route avoids arithmetic entirely but pays $\mathcal{O}(n_{\text{data}})$ in both gate count and subnormalisation. For the small instances in [1] the coherent-permutation route is genuinely better (row 1 of Table 5); asymptotically it is not. **We recommend the sparse-oracle route for any scaling study, and all asymptotic statements below assume it.**

5.4 Polynomial degree

With κ_{Q} from (47) and the accuracy allocation $\epsilon_{\text{inv}} = \epsilon/(5\kappa_{\text{Q}})$ from (44), the Chebyshev construction of [1, Appendix B] (reproduced in Appendix D) gives

$$b_{\text{ch}} = \kappa_{\text{Q}}^2 \log(\kappa_{\text{Q}}/\epsilon_{\text{inv}}), \qquad \boxed{d = \left\lceil \sqrt{b_{\text{ch}}}\,\log(4b_{\text{ch}}/\epsilon_{\text{inv}})\right\rceil = \mathcal{O}\big(\kappa_{\text{Q}}\log(\kappa_{\text{Q}}/\epsilon)\big).} \tag{54}$$

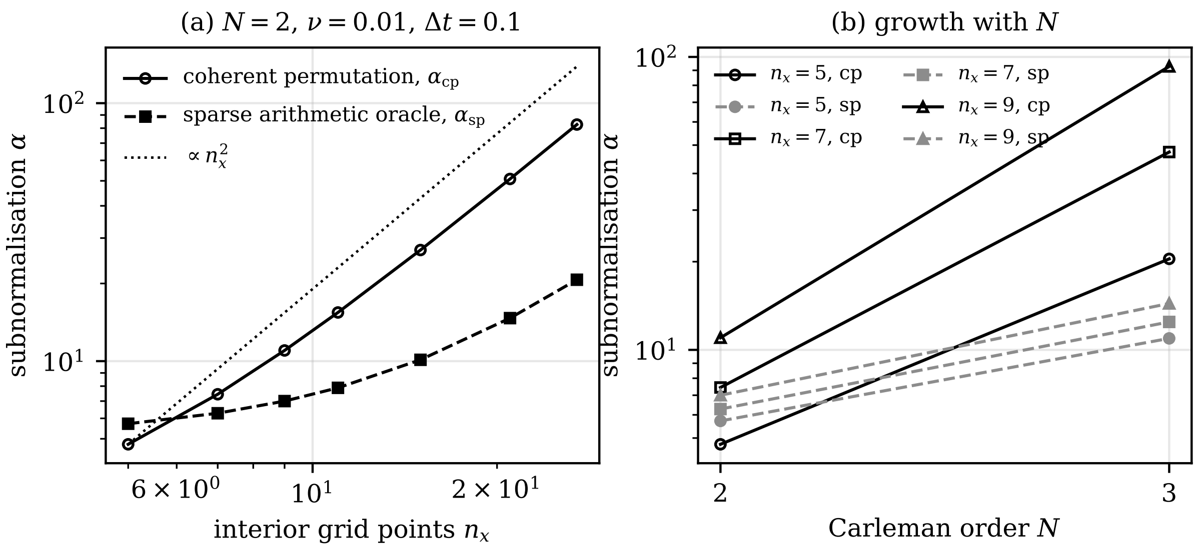

Figure 2: Growth of the block-encoding subnormalisation α, which sets the QSVT polynomial degree through $\kappa_{\mathrm{Q}} = \alpha/\sigma_{\min}$. (a) With grid size at fixed Carleman order $N = 2$: the coherent-permutation construction of [1] (circles) grows as $n_x^{N-1}\lambda_{\mathrm{cfl}} \propto n_x^2$ here because $\lambda_{\mathrm{cfl}} \propto n_x$ at fixed Δt, while the sparse arithmetic oracle (squares) stays nearly flat. (b) With Carleman order: the gap widens rapidly, reaching a factor 17 at $n_x = 15, N = 3$.

Remark 5.1 (A reproducibility gap in the source). The pairs (κ, d) reported in [1] are $(4, 117)$, $(8, 559)$ and $(12, 1275)$. Fitting a power law gives $d \propto \kappa^{2.1}$, whereas (54) predicts $d \propto \kappa$ up to logarithms. The discrepancy is resolved by noting that d also depends on ϵ_{inv}, which is not stated: inverting (54) shows the three cases correspond to $\epsilon_{\mathrm{inv}} \approx 10^{-2}$, 10^{-5} and 10^{-5} respectively. We record this because it means the published degrees cannot be compared with each other directly, and because any future scaling study should report ϵ_{inv} alongside κ.

5.5 Circuit depth

The QSVT circuit [1, Fig. 2] interleaves d applications of U_{L} (alternating with $U_{\mathsf{L}}^\dagger$) with $d+1$ projector-controlled phase shifts Π_{ϕ_i}, each a multi-controlled rotation on the flag register. Therefore

$$D_{\mathrm{QSVT}} \simeq d\,\big(D_{\mathrm{BE}} + D_{\Pi}\big), \tag{55}$$

where D_{BE} and D_{Π} are the two-qubit depths of one block encoding and one phase shift. We calibrate both against the four instances in [1, Tables 1,2,4]. Fitting the heavy-hex numbers gives

$$D_{\mathrm{BE}} \simeq c_{\mathrm{BE}}\, n_{\mathrm{data}}\, q_M^2, \qquad D_{\Pi} \simeq c_{\Pi}\, m_{\mathrm{flag}}^2, \qquad c_{\mathrm{BE}} \approx 10, \quad c_{\Pi} \approx 15. \tag{56}$$

The q_M^2 and m_{flag}^2 scalings are not assumed: the exponent on q_M is extracted from the two heat-equation rows, which differ only in q_M, via $\log(550/311)/\log(4/3) = 1.98$. Table 6 and Figure 3 show the model reproduces depths spanning a factor of 60 to within a factor of 1.6.

For the sparse-oracle route, D_{BE} is instead dominated by the arithmetic in O_{loc}: computing a column index requires $\mathcal{O}(1)$ additions and comparisons of q_M-bit integers, each of two-qubit depth $\mathcal{O}(q_M)$, plus a value-dependent rotation, so

$$D_{\mathrm{BE}}^{\mathrm{sp}} = \mathcal{O}\big(s\, q_M + q_M^2\big) = \mathcal{O}\big(N q_M + q_M^2\big)\,. \tag{57}$$

Table 6: Calibration of the depth model (55)–(56) against the heavy-hex two-qubit depths reported in [1]. D_{Π}^{obs} is inferred as $D_{\mathrm{QSVT}}/d - D_{\mathrm{BE}}$.

Instance	d	q_M	n_{data}	m_{flag}	D_{BE}	D_{Π}^{obs}	D_{QSVT}	model
Complex tridiagonal	117	3	6	4	283	204	57K	91K
Heat, $q_M = 3$	559	3	4	3	311	151	258K	277K
Heat, $q_M = 4$	1275	4	4	3	550	158	903K	988K
Burgers	559	5	14	5	5687	395	3.4M	2.2M

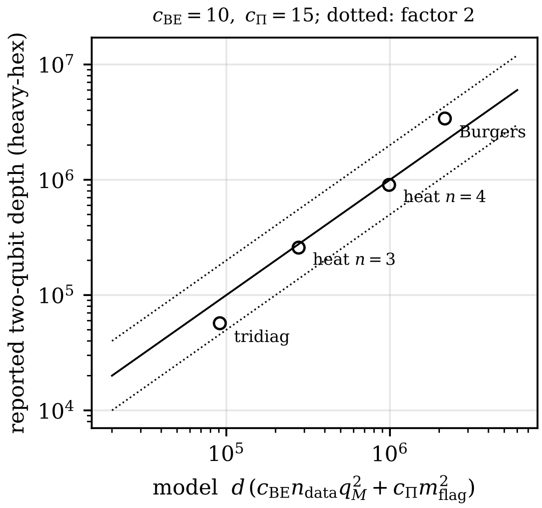

Figure 3: The depth model (55)–(56) against the four instances reported in [1]. Solid line: perfect agreement; dotted lines: a factor of two either way.

5.6 Success probability and the Carleman-block penalty

QSVT post-selection. Substituting $\beta = 1/(2\kappa_{\mathrm{Q}})$ into (27) and defining the *achieved* inverse norm $\kappa_{\mathrm{eff}} := \|(\mathsf{L}^\dagger/\alpha)^{-1}|b\rangle\|$, which lies in $[1, \kappa_{\mathrm{Q}}]$,

$$\boxed{p_s = \left(\frac{\kappa_{\mathrm{eff}}}{2\,\kappa_{\mathrm{Q}}^{\mathrm{design}}}\right)^2 \in \left[\frac{1}{4(\kappa_{\mathrm{Q}}^{\mathrm{design}})^2}, \frac{1}{4}\right].} \tag{58}$$

The upper bound 1/4 is reached when the design cut-off is tight. For the reference instance $\kappa_{\mathrm{eff}} = 4.83$ and $\kappa_{\mathrm{Q}}^{\mathrm{design}} = 8$, giving $p_s = 0.0911$ against the reported 0.0865 — agreement to 5%, which is the strongest single validation of our reconstruction. It also shows where the factor of three below the ideal 1/4 comes from: the conservative choice $\kappa = 8$ against an actual requirement of 5.5.

The Carleman block penalty. The solution state is the normalised Carleman vector $|\hat{y}\rangle = |(\hat{y}_1; \ldots; \hat{y}_N)\rangle$, but the physical field is only $\hat{y}_1$. Since $\|\hat{y}_j\| = \|\boldsymbol{u}\|^j$, writing $\gamma := \|\boldsymbol{u}\|_2$ the probability of finding the block register in state "1" is

$$\boxed{P_1 = \frac{\|\boldsymbol{u}\|^2}{\sum_{j=1}^N \|\boldsymbol{u}\|^{2j}} = \frac{\gamma^2}{\sum_{j=1}^N \gamma^{2j}} = \begin{cases} \dfrac{1-\gamma^2}{1-\gamma^{2N}}, & \gamma < 1, \\[2ex] \dfrac{\gamma^{-2(N-1)}(1-\gamma^{-2})}{1-\gamma^{-2N}}, & \gamma > 1. \end{cases}} \tag{59}$$

This is a genuine and, to our knowledge, undiscussed cost. If $\gamma > 1$ the Carleman vector is dominated by its *highest* block and P_1 decays geometrically, as $\gamma^{-2(N-1)}$. For the reference instance $\gamma^2 = \|\boldsymbol{u}_0\|^2 = 3$, giving $P_1 = 1/4$ at $N = 2$ but $P_1 = 2.8 \times 10^{-3}$ at $N = 6$ (Figure 1(b)).

The fix is to rescale $\boldsymbol{u} \to \boldsymbol{u}/\Gamma$ with $\Gamma > \gamma$, which by Remark 4.1 leaves the Carleman convergence ratio R untouched and makes $P_1 = 1 - \gamma^2/\Gamma^2 = \mathcal{O}(1)$. **We recommend normalising the discrete initial data to $\|\boldsymbol{u}_0\|_2 < 1$ before building the Carleman matrix**; it is free and it removes an exponential-in-N overhead.

5.7 The sequential-stepping obstruction, and the dilated system

Reference [1] advances one implicit step at a time: it solves $\mathsf{L}\boldsymbol{y}^{k+1} = \boldsymbol{y}^k$ for $k = 0, \dots, m-1$ with $m = T/\Delta t = 3$. In a state-vector simulation this is unproblematic, because the classical simulator can read out $\boldsymbol{y}^{k+1}$ and re-prepare it. On hardware it is not.

Proposition 5.2 (Sequential solves compound)**.** *Let each step succeed on post-selection with probability p_s. Then producing $|\boldsymbol{y}^m\rangle$ by chaining m QSVT solves requires $\mathcal{O}(p_s^{-m/2})$ applications of the chain even with optimal amplitude amplification.*

Sketch. Amplitude amplification of a state $U|0\rangle = \sqrt{p}|\text{good}\rangle + \dots$ to near-unit success requires $\mathcal{O}(p^{-1/2})$ applications of U, $U^\dagger$ and the reflection $2|0\rangle\langle 0| - \mathbb{I}$. To amplify step k alone one needs the reflection about *its* input state $U_{k-1} \cdots U_1|0\rangle$, which requires re-running the entire preceding chain. The costs therefore multiply rather than add, giving $\prod_k \mathcal{O}(p_s^{-1/2}) = \mathcal{O}(p_s^{-m/2})$. Amplifying the whole chain at once gives the same count, since $p_{\text{total}} = p_s^m$. $\square$

For the reference instance ($p_s = 0.087$, $m = 3$) this is a factor of 38 — an inconvenience. At $m = 300$ it is 10^{159}. Sequential stepping is therefore not merely inefficient but fatal, and any scaling claim must use the alternative.

The history-state formulation. Encode all time levels at once [7, 8, 9]. Introduce a time register of dimension $m + n_{\text{pad}} + 1$, where n_{pad} is a number of padding levels, and define the dilated matrix

$$\mathcal{L} = \begin{pmatrix} \mathbb{I} & & & & & & \\ -\mathbb{I} & \mathsf{L} & & & & & \\ & \ddots & \ddots & & & & \\ & & -\mathbb{I} & \mathsf{L} & & & \\ & & & -\mathbb{I} & \mathbb{I} & & \\ & & & & \ddots & \ddots & \\ & & & & & -\mathbb{I} & \mathbb{I} \end{pmatrix} \in \mathbb{R}^{(m+n_{\text{pad}}+1)D \times (m+n_{\text{pad}}+1)D}, \tag{60}$$

whose first row block imposes $\boldsymbol{y}^0 = \boldsymbol{y}_{\text{in}}$, whose next m row blocks impose the time stepping (22), and whose last n_{pad} row blocks simply copy $\boldsymbol{y}^m$ so as to raise its share of the total amplitude. Solving $\mathcal{L}\boldsymbol{Y} = (\boldsymbol{y}_{\text{in}}; 0; \dots; 0)$ in one QSVT call produces the *history state* $|\boldsymbol{Y}\rangle \propto \sum_k |k\rangle \otimes |\boldsymbol{y}^k\rangle$, from which $|\boldsymbol{y}^m\rangle$ is obtained by measuring the time register in the padding range.

Proposition 5.3 (Condition number of the dilated system)**.** *If all eigenvalues of A have non-positive real part, then*

$$\kappa(\mathcal{L}) \;\le\; \kappa_V(\mathsf{A})\,(m + n_{\text{pad}} + 1)\,\big(2 + \Delta t\|\mathsf{A}\|\big), \tag{61}$$

where $\kappa_V(\mathsf{A})$ is the eigenvector condition number, i.e. the condition number of the matrix V diagonalising $\mathsf{A} = V\Sigma V^{-1}$.

Proof. $\|\mathcal{L}\| \le 1 + \|\mathsf{L}\| \le 2 + \Delta t\|\mathsf{A}\|$ by the triangle inequality on the two nonzero block-diagonals. For the inverse, block forward-substitution gives $[\mathcal{L}^{-1}]_{k,j} = \mathsf{L}^{-(k-j)}$ for $k \ge j$ and 0 otherwise, so $\|\mathcal{L}^{-1}\| \le \sum_{k=0}^{m+n_{\text{pad}}} \|\mathsf{L}^{-k}\|$. Diagonalising $\mathsf{A} = V\Sigma V^{-1}$ gives $\|\mathsf{L}^{-k}\| \le \kappa_V \max_i |1 - \Delta t\mu_i|^{-k} \le \kappa_V$, since $\operatorname{Re}\mu_i \le 0$ implies $|1 - \Delta t\mu_i| \ge 1$. Summing the $m + n_{\text{pad}} + 1$ terms gives the claim. $\square$

Corollary 5.4 (The key cancellation). *With $n_{\text{pad}} = m$, and using $\|\mathsf{A}\| \simeq 4N\nu/\Delta x^2$ from (41) so that $\Delta t\|\mathsf{A}\| \simeq 4N\lambda_{\text{diff}}$,*

$$\kappa(\mathcal{L}) = \mathcal{O}\big(\kappa_V\, m\, N\lambda_{\text{diff}}\big) = \mathcal{O}\!\left(\kappa_V\, N\, \frac{T}{\Delta t} \cdot \frac{\Delta t}{\mathrm{Re}\Delta x^2}\right) = \boxed{\mathcal{O}\!\left(\frac{\kappa_V\, N\, T\, n_x^2}{\mathrm{Re}\, L^2}\right)} \tag{62}$$

which is independent of the time step Δt.

Corollary 5.4 is what makes the pathway analysable at all: the factors of Δt in the step count $m = T/\Delta t$ and in the diffusion number $\lambda_{\text{diff}} \propto \Delta t$ cancel exactly. Refining the time step costs nothing in the QSVT degree; it costs only in the size of the time register, i.e. $\log_2 m$ extra qubits. The physically meaningful cost is Tn_x^2/Re, which is the number of *diffusive* time scales resolved.

5.8 Success probability of the assembled algorithm

Collecting the three post-selections — QSVT flags, time register, and Carleman block — and writing $g := \max_k \|\boldsymbol{y}^k\|/\|\boldsymbol{y}^m\|$ for the norm decay of the trajectory,

$$\boxed{p_{\text{tot}} = \underbrace{p_s}_{\le 1/4} \times \underbrace{\frac{n_{\text{pad}}}{m + n_{\text{pad}}} \cdot \frac{1}{g^2}}_{\text{time register}} \times \underbrace{P_1}_{\text{Eq. (59)}}} \tag{63}$$

and with amplitude amplification the query cost multiplies by $\mathcal{O}(p_{\text{tot}}^{-1/2})$. With the recommendations of this section ($n_{\text{pad}} = m$, $\|\boldsymbol{u}_0\|_2 < 1$, tight κ) all three factors are $\mathcal{O}(1)$ and $p_{\text{tot}}^{-1/2} = \mathcal{O}(1)$; without them, each can be exponentially small.

5.9 Sample complexity

What must be measured, and how often, depends entirely on the deliverable. We distinguish three regimes; M_{shot} denotes the number of circuit repetitions.

1. **The state itself**, e.g. as input to a further quantum routine. Then $M_{\text{shot}} = \mathcal{O}(p_{\text{tot}}^{-1/2})$ coherent repetitions with amplitude amplification, i.e. $\mathcal{O}(1)$ with the recommendations above.

2. **A scalar observable** $\langle \mathcal{M} \rangle = \langle \hat{u}|\mathcal{M}|\hat{u}\rangle$ (drag, energy, a Fourier mode) to additive accuracy ϵ_{obs}. Naive sampling needs $M_{\text{shot}} = \mathcal{O}(\epsilon_{\text{obs}}^{-2} p_{\text{tot}}^{-1})$; amplitude estimation (Appendix F) reduces this to
$$M_{\text{shot}} = \mathcal{O}\!\left(\frac{1}{\epsilon_{\text{obs}}\sqrt{p_{\text{tot}}}}\right). \tag{64}$$
The normalisation $\|\hat{\boldsymbol{u}}(T)\|_2$, which the amplitude encoding does *not* carry, is obtained from the same estimator applied to the heralding flag at cost $\mathcal{O}(\epsilon_{\text{norm}}^{-1})$.

3. **The whole field**, i.e. all n_x amplitudes to relative ℓ^2 accuracy ϵ. Pure-state tomography requires
$$\boxed{M_{\text{shot}} = \mathcal{O}\!\left(\frac{n_x}{\epsilon^2\sqrt{p_{\text{tot}}}}\right),} \tag{65}$$
which is linear in the grid size and therefore destroys any exponential advantage on its own. This is the point [1] makes in its §2 and returns to in its discussion, and it is not avoidable: the amplitude encoding holds n_x real numbers in $\log_2 n_x$ qubits, and extracting n_x numbers takes $\Omega(n_x)$ measurements by an information-theoretic argument.

Regime 3 is what the challenge specification nominally asks for (an amplitude encoding of the solution), but read carefully it asks only for the *state*, which is regime 1. We flag the distinction because the two differ by a factor of n_x/ϵ^2 — for $n_x = 10^3$, $\epsilon = 10^{-2}$, that is 10^7.

5.10 Classical pre- and post-processing

The quantum circuit is not self-contained; three classical computations must accompany it.

1. **Estimating** $\sigma_{\min}(\mathsf{L})$ to fix the design κ. Reference [1] correctly notes that the $n_x^N \times n_x^N$ matrix must never be built: for the block-bidiagonal A the spectrum is the union of sums of j eigenvalues of F_1 (Appendix C), all available in closed form from (35). Cost: $\mathcal{O}(n_x)$ time, $\mathcal{O}(1)$ memory. For the non-normal case one may need Gershgorin bounds or a Lanczos iteration on the *structure*, still $\mathcal{O}(\mathrm{poly}(n_x, N))$.
2. **Computing the phase factors** $\vec{\phi} \in \mathbb{R}^{d+1}$. Naive root-finding on the Chebyshev expansion costs $\mathcal{O}(d^3)$ and loses precision; the optimisation-based methods of [17, 18] achieve $\mathcal{O}(d \operatorname{polylog} d)$ time and $\mathcal{O}(d)$ memory at machine precision. With $d \sim 10^6$ this is seconds, not a bottleneck — but it *is* a classical computation whose cost grows with κ_{Q}, and should be reported.
3. **Post-processing** the measurement record: negligible in regimes 1 and 2, $\mathcal{O}(n_x)$ in regime 3.

Classical memory for the whole pipeline is $\mathcal{O}(N + d)$: the $\mathcal{O}(N)$ distinct matrix values plus the phase list. *At no point is the $D \times D$ matrix instantiated*, and any implementation that does so has already lost.

5.11 Assembled scaling

We now combine everything. Assumptions, stated once: (a) sparse-oracle block encoding, (53); (b) high-order time integration, (42); (c) history-state formulation, (60); (d) initial data normalised to $\gamma < 1$; (e) $\mathrm{Re} < \pi/2$ so that N is finite; (f) the grid resolves the solution, $n_x = \Theta(\epsilon^{-1/2})$.

From Corollary 5.4, $\kappa_{\mathrm{Q}} = \mathcal{O}(\kappa_V N T n_x^2/\mathrm{Re})$. Substituting $n_x^2 = \Theta(\epsilon^{-1})$,

$$\kappa_{\mathrm{Q}} = \mathcal{O}\left(\frac{\kappa_V\, N\, T}{\mathrm{Re}\, \epsilon}\right), \qquad d = \widetilde{\mathcal{O}}(\kappa_{\mathrm{Q}}), \qquad q_M = \Theta(N \log_2 n_x) + \log_2(2m). \tag{66}$$

With $D_{\mathrm{BE}}^{\mathrm{sp}} = \mathcal{O}(N q_M + q_M^2) = \widetilde{\mathcal{O}}(N^2 \log^2 n_x)$ from (57), the sequential two-qubit depth is

$$\boxed{D_{\mathrm{QSVT}} = \widetilde{\mathcal{O}}\left(\frac{\kappa_V\, T}{\mathrm{Re}\, \epsilon}\, N^3 \log^2 n_x\, p_{\mathrm{tot}}^{-1/2}\right) \xrightarrow[p_{\mathrm{tot}} = \mathcal{O}(1)]{} \widetilde{\mathcal{O}}\left(\frac{\kappa_V\, T\, N^3}{\mathrm{Re}\, \epsilon}\right),} \tag{67}$$

with

$$N = \left\lceil \frac{\ln(1/\epsilon)}{\ln(\pi/(2\mathrm{Re}))} \right\rceil = \Theta\left(\frac{\log(1/\epsilon)}{\log(\pi/2\mathrm{Re})}\right), \qquad \#\mathrm{qubits} = \Theta\left(\frac{\log^2(1/\epsilon)}{\log(\pi/2\mathrm{Re})}\right). \tag{68}$$

Table 7 summarises, and Table 8 evaluates a concrete instance.

The comparison in Table 8 is stark: 10^{12} logical gates requiring gate infidelities of 10^{-12} (hence full fault tolerance, with a further 10^3–10^4 physical-qubit overhead per logical qubit) against one millisecond on a laptop. The gap is roughly 10^6 in operation count before error correction and 10^9–10^{10} after.

6 Comparison with classical solvers

6.1 What the classical state of the art costs

For one-dimensional Burgers with periodic or Dirichlet boundaries, the standard method is a pseudo-spectral discretisation (Fourier or Chebyshev) combined with an implicit–explicit or exponential time-differencing integrator: diffusion is treated exactly in spectral space, advection explicitly. Per time step the cost is dominated by two fast Fourier transforms, $\mathcal{O}(n_x \log n_x)$ arithmetic operations. The step size is limited by the advective Courant condition $\Delta t \lesssim \Delta x/U$, giving $m_{\mathrm{cl}} = \Theta(T n_x)$. Hence

$$\boxed{\mathrm{time}_{\mathrm{cl}} = \mathcal{O}(T\, n_x^2 \log n_x) = \widetilde{\mathcal{O}}\left(\frac{T}{\epsilon}\right), \qquad \mathrm{memory}_{\mathrm{cl}} = \mathcal{O}(n_x) = \mathcal{O}(\epsilon^{-1/2}),} \tag{69}$$

Table 7: Assembled resource scaling for the QSVT pathway applied to one-dimensional viscous Burgers, under assumptions (a)–(f) of Section 5.11. κ_V is the eigenvector condition number of the Carleman matrix (Proposition 5.3); it is the least controlled quantity in the table.

Resource	Scaling	Driver
Logical qubits	$\Theta(N \log n_x + \log m) = \Theta\big(\log^2(1/\epsilon)/\log \frac{\pi}{2\mathrm{Re}}\big)$	Carleman order × grid
Carleman order N	$\Theta\big(\log(1/\epsilon)/\log \frac{\pi}{2\mathrm{Re}}\big)$	diverges as $\mathrm{Re} \to \pi/2$
Grid points n_x	$\Theta\Big(\max(\mathrm{Re}, \epsilon^{-1/2})\Big)$	viscous layer, 2nd-order stencil
Time steps m	$\widetilde{\mathcal{O}}\big(TNn_x^2/\mathrm{Re}\big)$	only enters as $\log m$ qubits
Effective κ_{Q}	$\mathcal{O}\big(\kappa_V NTn_x^2/\mathrm{Re}\big) = \mathcal{O}(\kappa_V NT/(\mathrm{Re}\epsilon))$	dilated system
Polynomial degree d	$\widetilde{\mathcal{O}}(\kappa_{\mathrm{Q}})$	QSVT inversion
Block-encoding depth	$\mathcal{O}(Nq_M + q_M^2)$	sparse arithmetic oracle
Total depth	$\widetilde{\mathcal{O}}\big(\kappa_V TN^3\mathrm{Re}^{-1}\epsilon^{-1}\big)$	
Queries to $\mathcal{U}_0$	$\mathcal{O}(p_{\mathrm{tot}}^{-1/2}) = \mathcal{O}(1)$	amplitude amplification
Shots (state)	$\mathcal{O}(1)$	
Shots (observable)	$\mathcal{O}(\epsilon_{\mathrm{obs}}^{-1})$	amplitude estimation
Shots (full field)	$\mathcal{O}(n_x\epsilon^{-2})$	tomography
Classical time	$\widetilde{\mathcal{O}}(d) + \mathcal{O}(n_x)$	phase synthesis, $\sigma_{\min}$
Classical memory	$\mathcal{O}(N + d)$	never build the matrix

Table 8: A worked instance at the largest Reynolds number for which the Carleman series provably converges. Parameters: $\mathrm{Re} = 1$ (so $R_\star = 2/\pi = 0.637$), $\epsilon = 10^{-3}$, $T = 0.3$, $n_x = 2^7 = 128$, $\lambda_{\text{diff}} = 1$. Depths are logical two-qubit gates.

Quantity	Source	Value
Carleman order N	Eq. (34), $R_\star = 0.637$	16
System dimension D	Eq. (21), n_x^N	2^{112}
Matrix qubits q_M	Eq. (46)	112
Time steps m	$T/\Delta t$, $\Delta t = \mathrm{Re}\Delta x^2$	4915
Time register	$\lceil \log_2 2m \rceil$	14
Row sparsity s	$4N - 3$	61
Subnormalisation α_{sp}	Eq. (53)	2.0×10^3
Dilated $\kappa(\mathcal{L})$	Eq. (62), $\kappa_V = 1$	1.0×10^7
Polynomial degree d	Eq. (54)	2.3×10^8
Block-encoding depth	Eq. (57)	$\sim 5 \times 10^3$
Total depth	Eq. (55)	$\mathbf{1.3 \times 10^{12}}$
Total logical qubits	Eq. (45)	~ 135
Required gate infidelity	$\lesssim 1/D_{\mathrm{QSVT}}$	10^{-12}
Runtime at 100 ns/gate	—	~ 36 hours
Classical pseudo-spectral, same instance		
Arithmetic operations	$\mathcal{O}(m\, n_x \log n_x)$	4.4×10^6
Memory	$\mathcal{O}(n_x)$	1 kB
Runtime	—	~ 1 ms

where we used $n_x = \Theta(\epsilon^{-1/2})$ as in Section 5.11. Spectral methods in fact converge faster than second order for smooth solutions — the error decays exponentially in n_x, so $n_x = \Theta(\log(1/\epsilon))$ — which would make the classical cost $\widetilde{\mathcal{O}}(T\log^2(1/\epsilon))$ and the comparison even more lopsided. We deliberately use the pessimistic second-order figure (69) so that both sides are compared on the same discretisation.

6.2 Side by side

Table 9: Quantum (this analysis, Table 7) versus classical (Eq. (69)) for the d-dimensional analogue, with n_x points per dimension. $d = 1$ is Burgers; $d = 3$ would be Navier–Stokes. "Field" means the full solution vector is required; "observable" means only $\mathcal{O}(1)$ scalars are.

	Classical pseudo-spectral	Quantum Carleman + QSVT
Memory / qubits	$\mathcal{O}(n_x^d)$	$\mathcal{O}(Nd\log n_x)$
Work per unit time	$\mathcal{O}(n_x^{d+1}\log n_x)$	$\widetilde{\mathcal{O}}(\kappa_V N^3 n_x^2/\mathrm{Re})$
Dependence on ϵ	$\widetilde{\mathcal{O}}(\epsilon^{-1})$	$\widetilde{\mathcal{O}}(\epsilon^{-1}\log^3(1/\epsilon))$
Dependence on Re	$\mathcal{O}(\mathrm{Re}^{d+1})$ (resolution)	*no convergent regime for* $\mathrm{Re} > \pi/2$
Parallelisable	yes	no (sequential circuit depth)
Output	the field	a quantum state
Extra cost for the field	—	$\times\,\mathcal{O}(n_x^d\epsilon^{-2})$ tomography
Extra cost for an observable	$\mathcal{O}(n_x^d)$ (already have it)	$\mathcal{O}(\epsilon_{\mathrm{obs}}^{-1})$

Three conclusions follow from Table 9.

(i) In one dimension there is no advantage of any kind. Both sides scale as $\widetilde{\mathcal{O}}(\epsilon^{-1})$ in accuracy; the quantum side carries additional factors $\kappa_V N^3$, its depth is inherently sequential where the classical FFTs parallelise trivially, and it delivers a state rather than a field. The $\mathrm{Re} < \pi/2$ restriction excludes every physically interesting regime. The worked instance of Table 8 makes the gap concrete: nine to ten orders of magnitude.

(ii) The only structural quantum win is memory, and it survives. The qubit count $\mathcal{O}(Nd\log n_x)$ is genuinely exponentially smaller than $\mathcal{O}(n_x^d)$, and nothing in the analysis erodes it. A $d = 3$, $n_x = 1024$ problem needs 10^9 classical floating-point numbers versus roughly $Nd\log_2 n_x = 30N$ qubits. If a problem is memory-bound rather than time-bound, this is a real difference.

(iii) A crossover in work exists only for $d \geq 3$. Comparing the work rows, quantum wins when $n_x^{d+1}\log n_x \gtrsim \kappa_V N^3 n_x^2/\mathrm{Re}$, i.e. roughly $d \geq 2$ before constants and $d \geq 3$ after them. This is the familiar statement that quantum linear-algebra methods for PDEs can only pay off in high dimension. But the crossover is conditional on all of: $\mathrm{Re} = \mathcal{O}(1)$, observable-only readout, and $\kappa_V = \mathcal{O}(\mathrm{poly})$ — and the non-normality of Carleman matrices makes the last of these an open question.

6.3 The Cole–Hopf caveat

There is a further, uncomfortable point specific to Burgers. The Cole–Hopf transformation (Appendix H)

$$u = -2\nu\,\frac{\partial_x\phi}{\phi} = -2\nu\,\partial_x \ln\phi \tag{70}$$

maps (1) *exactly* onto the linear heat equation $\partial_t\phi = \nu\partial_{xx}\phi$. Burgers' equation is therefore not a genuinely nonlinear test problem at all: it is a linear problem in disguise, solvable classically in

$\mathcal{O}(n_x \log n_x)$ total (one FFT, a diagonal multiplication, one inverse FFT) for *any* T and *any* Re, with no time stepping whatsoever.

This does not invalidate [1], whose purpose is to demonstrate the pathway on a nonlinear-looking problem with a known answer — a reasonable choice of benchmark. But it does mean that a demonstration on Burgers cannot, by itself, be evidence that the pathway handles nonlinearity: the correct control experiment is a PDE with no linearising substitution, such as Korteweg–de Vries or Navier–Stokes. We note in passing that Cole–Hopf does not straightforwardly give a *quantum* algorithm either, since recovering $|u\rangle$ from $|\phi\rangle$ requires the nonlinear map $\phi \mapsto \partial_x \ln \phi$ on amplitudes, which is not implementable without tomography.

7 The Koopman–von Neumann variant

The challenge specification lists Koopman–von Neumann (KvN) linearisation as an alternative to Carleman. This section explains what changes. The short answer is that *stages 3–6 of the pathway are unchanged and the entire analysis of Section 5 transfers verbatim*; only stage 2 differs, and it differs in a way that trades one exponential for another.

7.1 Construction

Carleman lifts the state by taking *monomials of the solution*. KvN lifts it by taking *functions on the state space*. Start again from the semi-discrete system (12), written compactly as

$$\frac{\mathrm{d}\boldsymbol{u}}{\mathrm{d}t} = f(\boldsymbol{u}), \qquad f(\boldsymbol{u}) := \mathsf{F}_1\boldsymbol{u} + \mathsf{F}_2\boldsymbol{u}^{\otimes 2} \in \mathbb{R}^{n_x}. \tag{71}$$

Introduce an independent variable $\boldsymbol{\xi} \in \mathbb{R}^{n_x}$ ranging over the state space, and a density $\psi(t, \boldsymbol{\xi})$ on it. If ψ is transported by the flow of f, it obeys the *Liouville equation*

$$\boxed{\frac{\partial \psi}{\partial t} + \nabla_{\boldsymbol{\xi}} \cdot \big[f(\boldsymbol{\xi})\,\psi\big] = 0} \tag{72}$$

which is *linear in* ψ, exactly and with no truncation, for arbitrary nonlinearity f. Choosing $\psi(0, \boldsymbol{\xi}) = \delta(\boldsymbol{\xi} - \boldsymbol{u}_0)$ makes $\psi(t, \cdot)$ a delta function following the trajectory, and the solution is recovered as the first moment,

$$\boldsymbol{u}(t) = \int \boldsymbol{\xi}\, \psi(t, \boldsymbol{\xi})\, \mathrm{d}^{n_x}\xi = \langle \boldsymbol{\xi} \rangle_{\psi(t)}. \tag{73}$$

When $\nabla \cdot f = 0$ — true for Hamiltonian systems, *not* for Burgers, where $\nabla \cdot f = \operatorname{tr} \mathsf{F}_1 + \ldots \neq 0$ — equation (72) reduces to $\partial_t \psi = -f \cdot \nabla \psi$ and the generator $\hat{K} := -f(\boldsymbol{\xi}) \cdot \nabla_{\boldsymbol{\xi}}$ is anti-Hermitian, so $\mathrm{i}\hat{K}$ is Hermitian and the evolution is unitary. In general one keeps the full divergence form and $\hat{K}$ has an anti-Hermitian part plus a (bounded) Hermitian part.

7.2 Discretisation and what it costs

Discretise each of the n_x axes of $\boldsymbol{\xi}$ with M points, spacing $\Delta\xi$, using upwind or central differences for $\nabla_{\boldsymbol{\xi}}$. This yields exactly the same object the Carleman route produced — a linear ODE system

$$\frac{\mathrm{d}\boldsymbol{\psi}}{\mathrm{d}t} = \mathsf{K}\,\boldsymbol{\psi}, \qquad \boldsymbol{\psi} \in \mathbb{R}^{D_{\mathrm{KvN}}}, \qquad \boxed{D_{\mathrm{KvN}} = M^{n_x}} \tag{74}$$

which is then time-stepped implicitly, block-encoded and inverted by QSVT precisely as in Sections 3.3–3.5. Every formula of Section 5 applies with the substitutions

$$\mathsf{A} \to \mathsf{K}, \qquad D = n_x^N \;\to\; D_{\mathrm{KvN}} = M^{n_x}, \qquad q_M = N \log_2 n_x \;\to\; q_M^{\mathrm{KvN}} = n_x \log_2 M. \tag{75}$$

7.3 Where the two methods differ

Advantage: no linearisation error. Equation (72) is exact. The entire error source ε_{CL} of Section 4.3 — including the crippling restriction $\text{Re} < \pi/2$ — simply disappears, and is replaced by an ordinary finite-difference error $\mathcal{O}(\Delta\xi^r)$ in the phase-space discretisation. KvN works at any Reynolds number and any evolution time. This is a genuine and significant advantage.

Disadvantage: the state space is exponential in n_x, not in N. Compare the qubit counts:

$$q_M^{\text{Carleman}} = N \log_2 n_x \qquad \text{versus} \qquad q_M^{\text{KvN}} = n_x \log_2 M. \tag{76}$$

For a PDE this is decisive. With $n_x = 1024$ grid points and a modest $M = 2^8$ resolution per phase-space axis, Carleman at $N = 6$ needs 60 qubits and KvN needs 8192. Carleman's cost is exponential in the *accuracy parameter* N, which we get to choose; KvN's is exponential in the *problem size* n_x, which we do not. KvN is therefore attractive for small ODE systems — chemical kinetics, few-body dynamics — and unattractive for discretised PDEs.

Condition number. For the divergence-free case K is anti-Hermitian with purely imaginary eigenvalues $\mathrm{i}\omega$, so $\mathsf{L}_{\text{KvN}} = \mathbb{I} - \Delta t\mathsf{K}$ has singular values $|1 - \mathrm{i}\Delta t\omega| = \sqrt{1 + \Delta t^2\omega^2} \in [1, \sqrt{1 + \Delta t^2\|\mathsf{K}\|^2}]$. Hence

$$\kappa_{\text{Q}}^{\text{KvN}} = \frac{\alpha}{\sigma_{\min}} \simeq \alpha = s_{\text{KvN}} \|\mathsf{L}_{\text{KvN}}\|_{\max}, \qquad s_{\text{KvN}} = \mathcal{O}(n_x), \tag{77}$$

where the sparsity $s_{\text{KvN}} = 2n_x + 1$ comes from the n_x-dimensional stencil for $\nabla_{\boldsymbol{\xi}}$. Note $\sigma_{\min} \geq 1$ exactly, which is *better* than the Carleman case where non-normality can push it below 1 (we measured 0.872). But s is now $\mathcal{O}(n_x)$ rather than $\mathcal{O}(N)$, so the subnormalisation is worse by a factor n_x/N.

Initial state and readout. KvN's initial state $\delta(\boldsymbol{\xi} - \boldsymbol{u}_0)$ is a single computational basis state — trivial to prepare, and notably it does *not* need the amplitude oracle (3). But readout is harder: recovering $\boldsymbol{u}(T)$ requires estimating the n_x first moments (73), each an expectation value, at $\mathcal{O}(n_x/\epsilon_{\text{obs}})$ total cost with amplitude estimation. Against this, KvN offers a bonus Carleman cannot: because (72) is linear in ψ, one may superpose M_{ic} different initial conditions and obtain ensemble averages from a *single* solve, at no extra cost in M_{ic} [15]. For uncertainty quantification this is a real advantage.

7.4 Summary of the comparison

Table 10: Carleman versus Koopman–von Neumann as the linearisation stage of the same QSVT pathway. Stages 3–6 and hence the whole of Section 5 are common to both.

	Carleman (Section 3.2)	Koopman–von Neumann
Lifted variable	monomials $\boldsymbol{u}^{\otimes j}$, $j \leq N$	density $\psi(\boldsymbol{\xi})$ on state space
Exactness	truncated: error $\mathcal{O}(R^N)$	exact; only $\mathcal{O}(\Delta\xi^r)$ phase-space grid error
Convergence condition	$R_\star = 2\text{Re}/\pi < 1$	none
State dimension	n_x^N	M^{n_x}
Qubits q_M	$N \log_2 n_x$	$n_x \log_2 M$
Row sparsity s	$4N - 3$	$2n_x + 1$
$\sigma_{\min}(\mathsf{L})$	$\mathcal{O}(1)$, can be < 1 (non-normal)	≥ 1 exactly
Initial state	needs oracle (3)	computational basis state
Readout of $\boldsymbol{u}(T)$	first block, prob. P_1	n_x moment estimates
Many initial conditions	one solve each	one solve total
Verdict for a PDE	feasible qubit count, but fails for $\text{Re} \gtrsim 1.6$	works at any Re, but qubit count infeasible

Neither option is satisfactory for the challenge problem, and for opposite reasons. This is not an accident of these two particular methods: it reflects the lower bounds of Ameri, Carolan, Childs and Krovi [14], who show that simulating fluid dynamics at large Reynolds number is hard for quantum computers, and the limitation results of Lewis *et al.* [13] for turbulent and chaotic systems.

8 Discussion

8.1 What we found

The pathway of [1] is a correct and clearly presented construction, and its published gate-level numbers turn out to be an excellent calibration set: our independently derived formulas for the data-element count, the subnormalisation, the effective condition number and the success probability all reproduce them (Tables 2, 4, 6). Extrapolating those formulas, however, gives a discouraging picture, and it is worth being precise about *which* step is responsible for what.

- The **qubit count** is the one clean success: $\Theta(N \log n_x)$, exponentially better than classical memory, and nothing in the analysis erodes it.
- The **block encoding** as constructed is the largest avoidable cost: $\alpha \propto n_x^{N-1}$ (Eq. (49)) versus $\mathcal{O}(N^2)$ for an arithmetic sparse oracle (Eq. (53)), a factor of 17 already at $n_x = 15$, $N = 3$. This is fixable.
- The **sequential time stepping** is fatal as written (Proposition 5.2) but also fixable, via the dilated system; and the fix comes with a pleasant surprise, the Δt cancellation of Corollary 5.4.
- The **readout** is a hard information-theoretic barrier if the field is wanted, and a non-issue if only observables are.
- The **Carleman convergence condition** $\mathrm{Re} < \pi/2$ is *not* fixable within this framework. It is the binding constraint.

8.2 What would have to change

If one wanted to make this pathway competitive, the following are the specific targets, in decreasing order of importance.

1. **A Carleman analysis in better-adapted norms.** Our sharpened ratio $R_\star = 2\mathrm{Re}/\pi$ already improves on the naive grid-dependent estimate $R \propto \mathrm{Re}\, n_x^{3/2}$ by replacing the unweighted operator norm of F_2 with its restriction to the tensor manifold. Weighted-norm analyses in this spirit [12, 11] may push the threshold up, but they cannot remove it: at large Re the Burgers solution genuinely transfers energy to arbitrarily high monomial degree, and no truncation can represent that.
2. **Preconditioning.** κ_{Q} is dominated by α, and α by λ_{diff}. A block-encodable preconditioner for the diffusive part — for example diagonalising F_1 in the discrete sine basis, which is a quantum sine transform of depth $\mathcal{O}(q^2)$ — would remove the n_x^2/Re factor from Corollary 5.4 entirely. We regard this as the single most promising direction.
3. **Bounding κ_V.** The eigenvector condition number of the Carleman matrix appears in Proposition 5.3 and is, as far as we can tell, unbounded in the literature. Because A is block upper-bidiagonal with off-diagonal norm $\|\mathsf{F}_2\| \sim 1/\Delta x$, it is strongly non-normal and κ_V could plausibly grow with N or n_x. Any honest resource estimate must either bound it or measure it.

4. **Reformulating the deliverable.** A problem whose answer is a handful of functionals of $u(T,\cdot)$ — drag, dissipation rate, a spectrum — avoids the $\mathcal{O}(n_x\epsilon^{-2})$ tomography wall. A problem that wants the field does not.

5. **Choosing a genuinely nonlinear, genuinely high-dimensional target.** Burgers is linearisable by Cole–Hopf and one-dimensional; it can demonstrate that circuits compile but it cannot demonstrate advantage.

8.3 A note on how to report such estimates

Two small methodological points emerged that generalise beyond this paper. First, *state which condition number you mean*: $\alpha/\sigma_{\min}$ and $\sigma_{\max}/\sigma_{\min}$ differ by a factor of 3.5 in the published Burgers instance, and only the former sets the cost. Second, *report the polynomial accuracy ϵ_{inv} alongside κ and d*: without it, published (κ, d) pairs cannot be compared, as Section 5.4 shows.

8.4 Conclusion

For the one-dimensional viscous Burgers equation, the block-encoding + QSVT pathway delivers an exponential compression in memory and nothing else. Its sequential gate depth,

$$D_{\text{QSVT}} = \widetilde{\mathcal{O}}\big(\kappa_V\, T\, N^3\, \text{Re}^{-1}\, \epsilon^{-1}\big)\,,$$

matches the classical $\widetilde{\mathcal{O}}(T\epsilon^{-1})$ in its dependence on accuracy while carrying large extra factors; it returns a quantum state rather than a field; and it has a provable convergence regime only for $\text{Re} < \pi/2$. We would not present this as a route to advantage. We would present it as a fully costed construction that localises the obstruction precisely — and the localisation is itself the useful result, because three of the five obstructions listed in Section 8 are engineering problems with identifiable fixes, and only one, the Carleman convergence radius, is fundamental.

Reproducibility. Every numerical claim in this report is produced by three short Python scripts described in Appendix I; they reconstruct the matrices of [1] from its published equations and require only `numpy` and `scipy`.

A Finite differences, truncation error, and Gronwall

The stencils. Let u be four times continuously differentiable. Taylor-expanding about x_j with step Δx,

$$u_{j\pm1} = u(x_j \pm \Delta x) = u_j \pm \Delta x\, u_j' + \tfrac{\Delta x^2}{2} u_j'' \pm \tfrac{\Delta x^3}{6} u_j''' + \tfrac{\Delta x^4}{24} u_j'''' + \mathcal{O}(\Delta x^5). \tag{78}$$

Subtracting the two expansions cancels all even-order terms:

$$\frac{u_{j+1} - u_{j-1}}{2\Delta x} = u_j' + \frac{\Delta x^2}{6} u_j''' + \mathcal{O}(\Delta x^4), \tag{79}$$

and adding them cancels all odd-order terms:

$$\frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2} = u_j'' + \frac{\Delta x^2}{12} u_j'''' + \mathcal{O}(\Delta x^4). \tag{80}$$

Both are second-order accurate, which is what (10) claims. Substituting into (1) gives the local truncation error $\tau_j = \frac{\nu \Delta x^2}{12} u_j'''' - \frac{\Delta x^2}{6} u_j u_j''' + \mathcal{O}(\Delta x^4)$ quoted in Section 4.2.

Lemma A.1 (Gronwall)**.** *Let $e(t)$ satisfy $\dot e = g(t) + \Lambda e$ with $e(0) = 0$ and $\|g(t)\| \le \bar\tau$. Then $\|e(T)\| \le \bar\tau\,(e^{\Lambda T} - 1)/\Lambda$.*

Proof. By the variation-of-constants formula $e(T) = \int_0^T e^{\Lambda(T-s)} g(s)\,\mathrm{d}s$, so

$$\|e(T)\| \;\le\; \bar\tau \int_0^T e^{\Lambda(T-s)}\,\mathrm{d}s \;=\; \bar\tau\,\frac{e^{\Lambda T}-1}{\Lambda}. \qquad \square$$

Applying this with $g = \tau$ and Λ the one-sided Lipschitz constant of the semi-discrete right-hand side gives (30). The important qualitative point is that a local error $\mathcal{O}(\Delta x^2)$ becomes a global error $\mathcal{O}(\Delta x^2 e^{\Lambda T})$: nonlinearity makes errors grow exponentially in time, at a rate set by the velocity gradient.

Backward Euler error. For $\dot{\boldsymbol{y}} = \mathsf{A}\boldsymbol{y}$ the exact propagator over one step is $e^{\Delta t \mathsf{A}}$ and backward Euler applies $(\mathbb{I} - \Delta t\mathsf{A})^{-1}$. Expanding both,

$$(\mathbb{I}-\Delta t\mathsf{A})^{-1}-e^{\Delta t\mathsf{A}} = \big(\mathbb{I}+\Delta t\mathsf{A}+\Delta t^2\mathsf{A}^2+\dots\big)-\big(\mathbb{I}+\Delta t\mathsf{A}+\tfrac12\Delta t^2\mathsf{A}^2+\dots\big) = \tfrac12\Delta t^2\mathsf{A}^2+\mathcal{O}(\Delta t^3), \tag{81}$$

so the local error is $\frac12\Delta t^2\|\mathsf{A}\|^2$ and, over $m = T/\Delta t$ steps, the global error is $\frac12\Delta t\,T\|\mathsf{A}\|^2$, which is (40).

B Kronecker products and the Carleman blocks

Definition. For $\boldsymbol{a} \in \mathbb{R}^p$, $\boldsymbol{b} \in \mathbb{R}^r$ the Kronecker (tensor) product $\boldsymbol{a}\otimes\boldsymbol{b} \in \mathbb{R}^{pr}$ has entries $(\boldsymbol{a}\otimes\boldsymbol{b})_{ir+j} = a_i b_j$. For matrices, $(\mathsf{A}\otimes\mathsf{B})(\mathsf{C}\otimes\mathsf{D}) = (\mathsf{AC})\otimes(\mathsf{BD})$ whenever the products are defined — this mixed-product rule is the only identity we need. We write $\boldsymbol{u}^{\otimes j} := \boldsymbol{u}\otimes\cdots\otimes\boldsymbol{u}$ (j factors), a vector in $\mathbb{R}^{n_x^j}$ listing every degree-j monomial in the components of $\boldsymbol{u}$, with repetitions.

Lemma B.1 (Product rule for tensor powers)**.** *If $\boldsymbol{u} = \boldsymbol{u}(t)$ is differentiable then*

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}^{\otimes j} = \sum_{v=0}^{j-1} \boldsymbol{u}^{\otimes v}\otimes\dot{\boldsymbol{u}}\otimes\boldsymbol{u}^{\otimes(j-1-v)}. \tag{82}$$

Proof. The $(i_1,\dots,i_j)$ component of $\boldsymbol{u}^{\otimes j}$ is $u_{i_1}u_{i_2}\cdots u_{i_j}$. Differentiating a product of j factors gives j terms, the v-th of which differentiates the $(v+1)$-th factor. Reassembling into tensor notation gives the claim. $\square$

Substituting $\dot{\boldsymbol{u}} = \mathsf{F}_1\boldsymbol{u} + \mathsf{F}_2\boldsymbol{u}^{\otimes 2}$ into Lemma B.1 and using the mixed-product rule to pull the matrices out of the tensor product gives

$$\sum_v \boldsymbol{u}^{\otimes v}\otimes(\mathsf{F}_1\boldsymbol{u})\otimes\boldsymbol{u}^{\otimes(j-1-v)} = \Big[\sum_v \mathbb{I}^{\otimes v}\otimes\mathsf{F}_1\otimes\mathbb{I}^{\otimes(j-1-v)}\Big]\boldsymbol{u}^{\otimes j} = A_j^j\,\hat{y}_j, \tag{83}$$

$$\sum_v \boldsymbol{u}^{\otimes v}\otimes(\mathsf{F}_2\boldsymbol{u}^{\otimes 2})\otimes\boldsymbol{u}^{\otimes(j-1-v)} = \Big[\sum_v \mathbb{I}^{\otimes v}\otimes\mathsf{F}_2\otimes\mathbb{I}^{\otimes(j-1-v)}\Big]\boldsymbol{u}^{\otimes(j+1)} = A_{j+1}^j\,\hat{y}_{j+1}, \tag{84}$$

which are exactly (18)–(19). Note the dimensions: A_j^j is square ($n_x^j \times n_x^j$) because F_1 is, while A_{j+1}^j is *rectangular* ($n_x^j \times n_x^{j+1}$) because F_2 is $n_x \times n_x^2$. That rectangularity is what breaks the constant-diagonal structure exploited by the coherent-permutation block encoding, and hence the origin of the $\Theta(n_x^{N-1})$ data-element count in Section 5.2.

C Spectra of the discrete operators

The diffusion matrix. $\mathsf{F}_1 = \lambda_1\mathrm{tridiag}(1,-2,1)$ with Dirichlet boundaries has the classical eigenpairs

$$\mu_k = -4\lambda_1\sin^2\!\Big(\frac{k\pi}{2(n_x+1)}\Big), \qquad (w_k)_j = \sin\!\Big(\frac{jk\pi}{n_x+1}\Big), \qquad k = 1,\dots,n_x, \tag{85}$$

verified by substituting w_k into the difference equation and using $\sin(\theta \pm \phi)$ addition formulas. With $\lambda_1 = \nu/\Delta x^2$ this is (35). The eigenvectors are orthogonal, so F_1 is normal and $\kappa_V(\mathsf{F}_1) = 1$; this is *not* inherited by A. For periodic boundaries the eigenvalues become $-4\lambda_1 \sin^2(k\pi/n_x)$ with $k = 0, \ldots, n_x - 1$, which includes a zero mode $k = 0$ — and hence $\operatorname{Re}\mu_1 = 0$, formally invalidating the Carleman criterion (33). The zero mode is the constant function, which is conserved by Burgers; restricting to the mean-zero subspace (as the initial data $\sin(2\pi x)$ does) restores $|\operatorname{Re}\mu_1| = 4\lambda_1 \sin^2(\pi/n_x) \to 4\pi^2\nu/L^2$, four times the Dirichlet value. This is one of the two places where periodicity changes a constant.

The Carleman matrix. Since A in (20) is block upper-triangular, its spectrum is the union of the spectra of its diagonal blocks A_j^j. By the mixed-product rule, if $\mathsf{F}_1 w_k = \mu_k w_k$ then

$$A_j^j\big(w_{k_1} \otimes \cdots \otimes w_{k_j}\big) = (\mu_{k_1} + \cdots + \mu_{k_j})\,\big(w_{k_1} \otimes \cdots \otimes w_{k_j}\big), \tag{86}$$

so the eigenvalues of A are all j-fold sums of eigenvalues of F_1, $j = 1, \ldots, N$. Consequently

$$|\operatorname{Re}\mu|_{\min}(\mathsf{A}) = |\mu_1(\mathsf{F}_1)| \to \frac{\pi^2\nu}{L^2}, \qquad |\operatorname{Re}\mu|_{\max}(\mathsf{A}) = N|\mu_{n_x}(\mathsf{F}_1)| \to \frac{4N\nu}{\Delta x^2}, \tag{87}$$

which is what (41) uses, and all eigenvalues are real and negative — justifying the hypothesis of Proposition 5.3. The *eigenvectors*, however, are not orthogonal: the off-diagonal blocks A_{j+1}^j tilt them, by an amount controlled by $\|A_{j+1}^j\|/\big(\text{eigenvalue gap}\big)$. This is the source of $\kappa_V(\mathsf{A}) > 1$ and of the measured $\sigma_{\min}(\mathsf{L}) = 0.872 < 1$.

D Quantum signal processing, QSVT, and the inverse polynomial

QSP. Quantum signal processing is the observation that interleaving a fixed single-qubit rotation depending on a parameter a with tunable Z-rotations realises a polynomial in a. In the W_X convention [1, Appendix A.1] the signal operator is $W_X(a) = \begin{pmatrix} a & i\sqrt{1-a^2} \\ i\sqrt{1-a^2} & a \end{pmatrix}$ and, for a phase list $\vec{\phi}' \in \mathbb{R}^{d+1}$,

$$U_{W_X}(\vec{\phi}') = e^{i\phi_0' Z} \prod_{k=1}^{d} W_X(a)\, e^{i\phi_k' Z}, \qquad \operatorname{Poly}(a) = \langle + | U_{W_X}(\vec{\phi}') | + \rangle. \tag{88}$$

The achievable Poly are exactly the real polynomials of degree $\le d$, parity $d \bmod 2$, and bounded by 1 on $[-1, 1]$ [1, Theorem 1]. The boundedness constraint is why the normalisation β appears: $1/a$ is unbounded, so one can only realise β/a with β small enough that the result stays in $[-1, 1]$, which forces $\beta \le 1/(2\kappa)$ when the domain reaches down to $a = 1/\kappa$.

QSVT. QSVT lifts this from a scalar a to the singular values of a block-encoded matrix. Replacing $W_X(a)$ by U_{L} and the Z-rotations by projector-controlled phase shifts Π_ϕ (multi-controlled rotations that act only when the flag register is $|0\rangle$), the resulting circuit has top-left block $\sum_k \operatorname{Poly}(\sigma_k)|w_k\rangle\langle v_k|$ [1, Theorem 2]. The crucial feature is that this is achieved *without ever computing the singular values.*

The inverse polynomial. Following [1, Appendix B], one approximates

$$f(a) = \frac{1 - (1 - a^2)^b}{a}, \qquad b_{\text{ch}} = \kappa^2 \log(\kappa/\epsilon_{\text{inv}}), \tag{89}$$

which is ϵ_{inv}-close to $1/a$ outside $[-1/\kappa, 1/\kappa]$ and, unlike $1/a$, bounded. Expanding f in Chebyshev polynomials T_i of the first kind and truncating gives

$$P^{1/a}_{2\epsilon_{\text{inv}},\kappa}(a) = 4\sum_{j=0}^{d}(-1)^j\left[2^{-2b_{\text{ch}}}\sum_{i=j+1}^{b_{\text{ch}}}\binom{2b_{\text{ch}}}{b_{\text{ch}}+i}\right]T_{2j+1}(a), \qquad d = \left\lceil\sqrt{b_{\text{ch}}}\,\log(4b_{\text{ch}}/\epsilon_{\text{inv}})\right\rceil, \tag{90}$$

which is Eq. (54). The asymptotic form $d = \mathcal{O}(\kappa\log(\kappa/\epsilon))$ follows since $\sqrt{b_{\text{ch}}} = \kappa\sqrt{\log(\kappa/\epsilon_{\text{inv}})}$.

E Block encoding: two constructions

What it is. A block encoding, Eq. (24), hides a non-unitary L inside a unitary by enlarging the Hilbert space. Acting on $|0^{m_{\text{flag}}}\rangle|b\rangle$ and post-selecting the flag register on $|0^{m_{\text{flag}}}\rangle$ implements L/α; the failure branch is the "$*$" garbage. The subnormalisation $\alpha \geq \|\mathsf{L}\|_2$ is unavoidable because a submatrix of a unitary has singular values at most 1.

Construction 1: sparse-access oracle. Suppose L has at most s nonzeros per row and we are given two oracles,

$$O_{\text{loc}}|r\rangle|\ell\rangle = |r\rangle|c(r,\ell)\rangle, \qquad O_{\mathsf{L}}|r\rangle|c\rangle|0\rangle = |r\rangle|c\rangle\left(\tfrac{\mathsf{L}_{rc}}{\|\mathsf{L}\|_{\max}}|0\rangle + *|1\rangle\right), \tag{91}$$

where $c(r,\ell)$ is the column of the ℓ-th nonzero in row r. The standard construction (diffusion on the ℓ register, apply O_{L}, apply O_{loc}, diffusion again) yields a block encoding with

$$\alpha_{\text{sp}} = s\,\|\mathsf{L}\|_{\max}, \tag{92}$$

which is Eq. (52). For the Carleman matrix both oracles are computable arithmetically from the closed forms (18)–(19): given a row index, one extracts the block index j and the multi-index by integer division by powers of n_x, then adds or subtracts n_x^v. The gate cost is $\mathcal{O}(1)$ ripple-carry adders and comparators on q_M-bit registers, giving Eq. (57).

Construction 2: coherent permutation. The scheme of [2] used in [1] instead assumes the distinct values of L repeat along constant diagonals. It loads the list of distinct values into a data register with a state-preparation unitary (PREP), uses shift operators $O_s(k,\text{diag})$ to move the k-th value onto its diagonal, uses delete/insert operators to fix up boundary exceptions, and un-prepares. Then $\alpha = \|v_{\text{data}}\|_1$. Its virtue is that it needs no arithmetic and compiles to nearest-neighbour circuits; its cost is that both the gate count and α grow with the number n_{data} of distinct (value, offset) pairs, which for Carleman matrices is $\Theta(n_x^{N-1})$ because the blocks A^j_{j+1} are rectangular (Appendix B).

F Amplitude amplification and estimation

Amplification. If $U|0\rangle = \sqrt{p}|\text{good}\rangle + \sqrt{1-p}|\text{bad}\rangle$, then alternating the reflections $R_{\text{good}} = \mathbb{I} - 2\Pi_{\text{good}}$ and $R_0 = 2|0\rangle\langle 0| - \mathbb{I}$ rotates the state within the two-dimensional span by an angle $2\arcsin\sqrt{p}$ per iteration. After $\mathcal{O}(1/\sqrt{p})$ iterations the amplitude on $|\text{good}\rangle$ is $\Theta(1)$. This is the standard quadratic speedup over repeat-until-success, which would need $\mathcal{O}(1/p)$ trials, and is why all our success-probability penalties enter as $p^{-1/2}$. Fixed-point variants [19] avoid overshoot at the cost of a $\log(1/\delta)$ factor, where $1-\delta$ is the guaranteed success probability.

Estimation. Applying phase estimation to the amplification operator gives $\hat{p}$ with $|\hat{p}-p| \leq \epsilon_{\text{obs}}$ using $\mathcal{O}(1/\epsilon_{\text{obs}})$ applications of U, versus $\mathcal{O}(1/\epsilon^2_{\text{obs}})$ for naive sampling. This is the source of the two rates quoted in Section 5.9.

Why chaining breaks it. Amplifying step k of a chain requires R_0 about *its* input state, i.e. the reflection $2|\psi_{k-1}\rangle\langle\psi_{k-1}| - \mathbb{I}$ where $|\psi_{k-1}\rangle = U_{k-1}\cdots U_1|0\rangle$. Implementing it requires $U_{k-1}\cdots U_1$ and its inverse. Recursing gives the multiplicative $\prod_k p_k^{-1/2}$ of Proposition 5.2.

G Perturbation bound: why κ_q amplifies

Lemma G.1. *Let* $\mathsf{L}\boldsymbol{y} = \boldsymbol{b}$ *and* $(\mathsf{L} + \delta\mathsf{L})(\boldsymbol{y} + \delta\boldsymbol{y}) = \boldsymbol{b} + \delta\boldsymbol{b}$*, with* $\|\mathsf{L}^{-1}\|\|\delta\mathsf{L}\| < 1$*. Then*

$$\frac{\|\delta\boldsymbol{y}\|}{\|\boldsymbol{y}\|} \le \frac{\kappa}{1 - \kappa\,\|\delta\mathsf{L}\|/\|\mathsf{L}\|}\left(\frac{\|\delta\mathsf{L}\|}{\|\mathsf{L}\|} + \frac{\|\delta\boldsymbol{b}\|}{\|\boldsymbol{b}\|}\right), \qquad \kappa := \|\mathsf{L}\|\|\mathsf{L}^{-1}\|. \tag{93}$$

Proof. Subtracting the two systems, $\mathsf{L}\,\delta\boldsymbol{y} = \delta\boldsymbol{b} - \delta\mathsf{L}\,(\boldsymbol{y} + \delta\boldsymbol{y})$, so $\|\delta\boldsymbol{y}\| \le \|\mathsf{L}^{-1}\|\big(\|\delta\boldsymbol{b}\| + \|\delta\mathsf{L}\|\|\boldsymbol{y}\| + \|\delta\mathsf{L}\|\|\delta\boldsymbol{y}\|\big)$. Collecting $\|\delta\boldsymbol{y}\|$ on the left and using $\|\boldsymbol{b}\| \le \|\mathsf{L}\|\|\boldsymbol{y}\|$ gives the result. □

In our setting $\|\mathsf{L}\|$ is replaced by α (the block encoding realises L/α, and errors are measured relative to that), so the amplification factor is $\kappa_{\mathrm{Q}} = \alpha/\sigma_{\min}$ rather than $\sigma_{\max}/\sigma_{\min}$ — the same quantity identified in Section 5.2. This is why the four middle rows of Table 3 must be controlled to $\epsilon/\kappa_{\mathrm{Q}}$: they perturb the *system*, whereas discretisation errors perturb the *model* and are not amplified.

H The Cole–Hopf transformation

Let $\phi(t,x) > 0$ solve the heat equation $\partial_t\phi = \nu\partial_{xx}\phi$ and set $u := -2\nu\,\partial_x \ln\phi = -2\nu\,\phi_x/\phi$. Then

$$u_t = -2\nu\frac{\phi_{xt}\phi - \phi_x\phi_t}{\phi^2} = -2\nu^2\frac{\phi_{xxx}\phi - \phi_x\phi_{xx}}{\phi^2}, \tag{94}$$

$$u\,u_x = \Big(-2\nu\frac{\phi_x}{\phi}\Big)\Big(-2\nu\frac{\phi_{xx}\phi - \phi_x^2}{\phi^2}\Big) = 4\nu^2\frac{\phi_x\phi_{xx}\phi - \phi_x^3}{\phi^3}, \tag{95}$$

$$\nu u_{xx} = -2\nu^2\frac{\phi_{xxx}\phi^2 - 3\phi_x\phi_{xx}\phi + 2\phi_x^3}{\phi^3}. \tag{96}$$

Adding the first two and comparing with the third shows $u_t + uu_x - \nu u_{xx} = 0$ identically. So every solution of the linear heat equation generates a solution of Burgers, and (since $\phi = \exp(-\frac{1}{2\nu}\int u)$ inverts the map) conversely. Burgers is therefore exactly linearisable, which is the caveat discussed in Section 6. The Korteweg–de Vries equation admits no such substitution — it is integrable by the inverse scattering transform, which is not a change of dependent variable — and is for that reason the more honest test case.

I Numerical validation

All numbers attributed to "this work" are produced by three scripts, each depending only on `numpy` and `scipy`.

1. `verify.py` reconstructs the Burgers instance of [1, §4.3] from Eqs. (14), (18)–(19) and (22), and checks: $D = 30$; $q_M = 5$; the 14 distinct (value, offset) pairs and their numerical values against [1, Table 3]; $\alpha = 4.760$; $\sigma_{\min} = 0.872$; the Carleman ratio $R = 76.2$; the block-1 probability P_1; and the success probability $p_s = 0.0911$ against the reported 0.0865.

2. `scaling.py` builds L for a grid of (n_x, N) values using sparse Kronecker products, and measures n_{data}, α, the row sparsity, and $\kappa(\mathsf{L})$. It verifies the α formula (49) to three decimal places over $\Delta t \in [0.01, 4]$ and establishes the $\Theta(n_x^{N-1})$ growth of n_{data}.

3. `kappa.py` settles the κ_{Q} question of Table 4 by recomputing α, $\sigma_{\max}$, $\sigma_{\min}$ for all three worked examples of [1], and produces Table 5.

Selected raw output, for the reference instance of Table 2:

```
Dim(A) = 30   (paper: 30)   matrix qubits = 5 (paper: 5)
#(value,offset) pairs = 14   (paper: 14 data elements -> 4 data qubits)
distinct nonzero values of L:  -0.3000 x4  -0.0360 x88  +0.3000 x4
                               +1.0720 x5  +1.1440 x25
alpha (sum |distinct data values|) = 4.7600
sigma_max(L)=1.3702  sigma_min(L)=0.8722  kappa(L)=1.5709
R = ||u0|| ||F2|| / |Re lam1| = 76.180    (need R<1 for the Liu et al. bound)
kappa_eff = 4.8278; with kappa_design=8: ps=0.09105 (paper 0.0865)
```

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